What Moves the Classroom? A Bayesian Latent-State Model of Pedagogy and 21st-Century Skills in Whole-Classroom Discourse

This Supplementary Material covers every model from first principles: motivation, mathematical formulation, prior choices, forward algorithm derivation, worked numerical examples, and code. Updated version of this Supplementary Material can be found on https://github.com/drake1729/LAK27

October 9, 2026

Contents

1 What Are We Modelling? 4
1.1 The Problem . . . 4
1.2 Data at a Glance . . . 4
1.3 Why These Three Dimensions . . . 4
1.4 The Covariate Vector . . . 5
1.5 The Three Models and Their Trade-Offs . . . 5

2 Model 1: BLFM — Bayesian Latent Factor Model 6
2.1 Core Question . . . 6
2.2 The Exchangeability Assumption . . . 6
2.3 The Sufficient Statistic: Episode-Level Aggregation . . . 6
2.4 Model Specification . . . 7
2.4.1 Step 1 — Automatic Relevance Determination (ARD) Prior . . . 7
2.4.2 Step 2 — Intercept and Random Effects . . . 7
2.4.3 Step 3 — Predicted Episode Rate . . . 7
2.4.4 Step 4 — Beta Regression Likelihood . . . 8
2.5 Full Generative Story . . . 8
2.6 Worked Numerical Example . . . 8
2.7 Code . . . 9
2.7.1 Key Implementation Details Explained . . . 10
2.8 Running MCMC . . . 10
2.9 What BLFM Cannot Answer . . . 11

3 Model 2: PST — Probabilistic Skill Tracing (Sequential BKT) 11
3.1 Core Question . . . 11
3.2 Hidden Markov Model (HMM) Primer . . . 11
3.3 PST Parameters . . . 11
3.3.1 The Transition Matrix . . . 12
3.3.2 Emission Probabilities . . . 12
3.3.3 Why Are p_S and p_G Fixed? . . . 13
3.4 The Forward Algorithm: Step-by-Step Derivation . . . 13
3.4.1 Why Not Enumerate All State Sequences? . . . 13
3.4.2 Notation . . . 13
3.4.3 Initialisation ($t = 1$) . . . 13
3.4.4 Recursion ($t \geq 2$): Predict–Update . . . 14
3.4.5 Log-Likelihood . . . 14
3.5 Worked Numerical Forward Pass . . . 14
3.6 Prior Choices Explained . . . 15
3.7 Key Result: Regime Persistence . . . 15
3.8 Code . . . 15
3.8.1 Two Technical Details . . . 17
3.9 Why PST's AUC is Misleading . . . 17

4 Model 3: IO-BKT — Input-Output Bayesian Knowledge Tracing 18
4.1 Core Question . . . 18

4.2 The Key Innovation: Content-Driven Transitions . . . 18
4.3 Full Model Specification . . . 19
4.3.1 Priors . . . 19
4.3.2 Emission Model . . . 19
4.3.3 Forward Algorithm with Time-Varying Transitions . . . 20
4.4 Worked Numerical Example: Transition Probabilities . . . 20
4.5 Interpreting Table II: The Posterior Coefficients . . . 20
4.6 Code . . . 21
4.6.1 The `einsum` Operation Decoded . . . 23
4.6.2 MCMC Configuration . . . 23
4.7 Label Switching: Detection and Correction . . . 24
4.8 The Full-Length Normalisation (Tmax Extension) . . . 25

5 Experimental Setup 26

6 Results and Discussions 26
6.1 Baselines . . . 26
6.2 What Each AUC Gap Means . . . 27
6.3 PST: Regime Persistence Finding . . . 28
6.4 The Three-Stage AI Coaching Pipeline . . . 28
6.5 Summary: The Three Models Side by Side . . . 30
6.6 Conclusion . . . 30
6.7 Knowledge Change Curves . . . 31
6.7.1 Motivation . . . 31
6.7.2 Notation . . . 31
6.7.3 Episode Opening Baseline . . . 31
6.7.4 Feature Occurrence Indexing . . . 32
6.7.5 Per-Occurrence Change from Opening . . . 32
6.7.6 The Knowledge Change Curve . . . 32
6.7.7 Axis Definitions . . . 33
6.7.8 Why $P(M_t{=}1)$ and Not O_t . . . 33
6.7.9 Comparison with Scarlatos et al. . . . 33
6.7.10 Interpretation of Knowledge Change Curves . . . 33
6.8 Teacher → Student Lag Analysis . . . 34
6.8.1 Why "Lag" Analysis . . . 34
6.8.2 Constructing Matched Pairs . . . 34
6.8.3 EP-Conditional Student 21CS Rate . . . 35
6.8.4 EP-Conditional Student Cognitive Process Level . . . 35
6.8.5 Lesson-Phase Analysis via Global Encoding . . . 35
6.8.6 Teacher → Student Lag Analysis plots . . . 36
6.9 Bloom's Cognitive Process Trajectory Across Lesson Time . . . 38
6.9.1 Method . . . 38
6.9.2 Bloom's CP learning curve and 21CS activation across the lesson time plots . . . 39
6.10 Expert Validation . . . 39
6.11 Thematic Analysis . . . 41
6.12 Teacher Productive State . . . 42

6.12.1 Motivation 42
6.12.2 Teacher-Level Results 43
6.12.3 Utility for Learning Analytics and Teacher Profiling 44

1 What Are We Modelling?

1.1 The Problem

Teaching is sequential: a teacher speaks, students respond, the cognitive climate of the room shifts. The paper asks: *can we track whether a classroom is in a "productive discourse regime", one that activates 21st-Century Skills (21CS), utterance by utterance?*

This is discourse Pedagogical State Tracking. It adapts Bayesian Knowledge Tracing (BKT), originally designed for tracking a single student's mastery of a single topic, to the whole-classroom level.

1.2 Data at a Glance

Property	**Value**	**Note**
Episodes E	49 (48 usable)	31 Science, 18 Math; Grades 6–12
Total lines	19,181	Median episode: 384 lines
Mean duration	28.7 min	SD 4.8; range 12–39 min
Source	Indian classroom (48/49)	1 episode from vignette named, 'Volcanoes? Here?'
Feature	**Dimensions**	**Description**
EP (Educational Practice)	3	Behaviourism, Cognitivism, Constructivism
KD (Knowledge Dimension)	4	Factual, Conceptual, Procedural, Metacognitive
CP (Cognitive Process)	6	Remembering, Understanding, Applying, Analysing, Evaluating, Creating
Outcome		
$O_t \in \{0, 1\}$	1	1 if any 21CS active at line t (prevalence 0.482)
21CS	4	Problem Solving, Creativity, Collaborative Learning, Critical Thinking

1.3 Why These Three Dimensions

Instructional design is commonly described through three components: the strategies a teacher uses, the learner, and the content Ertmer2013. Our covariates $\mathbf{z}_t$ map directly onto these (Fig. 1): strategies to EP (learning theories), the learner's cognitive engagement to CP, and content to KD, thereby achieving line-level coverage. Subject and grade, the remaining content and learner factors, enter as the Subject×Grade random effect u_e and receive coarser, episode-level coverage through the Subject×Grade hierarchical random effect $u_e \sim \mathcal{N}(0, \sigma^2_{\text{cell}})$ rather than as line-level covariates. Ertmer2013 argue that no learning theory is best in general; each suits tasks of different complexity Ertmer2013. Our coefficients, therefore, describe association with 21CS activation in particular, not the overall quality of a teaching move.

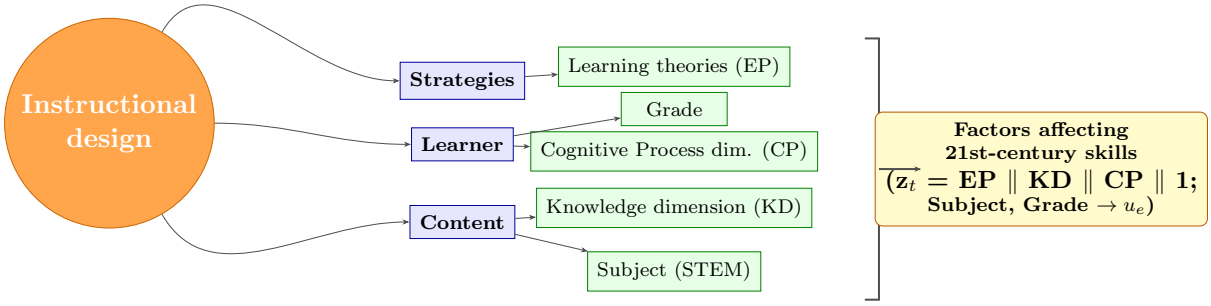


Figure 1: Instructional-design framework operationalised in this paper. Factors affecting 21st-century skills.

1.4 The Covariate Vector

At every line t, we have a 14-dimensional binary vector (13 content features + 1 intercept):

$$\mathbf{z}_t = [\underbrace{\text{Beh, Cog, Con}}_{\text{3 EP}} \parallel \underbrace{\text{Fac, Con, Pro, Met}}_{\text{4 KD}} \parallel \underbrace{\text{Rem, Und, App, Ana, Eva, Cre}}_{\text{6 CP}} \parallel \underbrace{1}_{1}] \in \{0,1\}^{13} \times \{1\} \tag{1}$$

1.5 The Three Models and Their Trade-Offs

Model	Core Assumption	Temporal?	Content-Driven?	AUC
BLFM	Episode = bag of lines	**No**	Yes (mean)	0.772
PST	HMM, fixed transitions	**Yes**	**No**	0.928*
IO-BKT	HMM, content-driven trans.	**Yes**	**Yes**	**0.873**

* Inflated by autocorrelation, explained in §3.9.

The models are not competitors: they *answer different questions* and form a deliberate complexity ladder. We start with the simplest.

2 Model 1: BLFM — Bayesian Latent Factor Model

2.1 Core Question

BLFM Answers

Does the average mix of pedagogical moves in an episode predict how often 21CS will be active, regardless of the order in which those moves appear?

2.2 The Exchangeability Assumption

BLFM makes one fundamental assumption: lines within an episode are **exchangeable** — like balls drawn from a bag. Shuffling the lines in any order produces the same prediction.

This is, at first, startling. Why would we ignore temporal order?

Why Exchangeability Is a Reasonable Starting Point

PST (Model 2) estimates $\hat{p}_L = 0.073$: the classroom changes regime roughly *once every 14 lines*. An episode with 384 lines sees only $\sim$27 transitions. The classroom spends most of its time locked in one regime. Therefore, the *fraction* of lines spent in each regime (captured by the mean feature vector) is a reasonable proxy for the dominant regime quality. Order only matters at the few transition moments.

2.3 The Sufficient Statistic: Episode-Level Aggregation

For episode e with T_e lines, BLFM collapses the full sequence into two summaries:

$$\bar{\mathbf{x}}_e = \frac{1}{T_e}\sum_{t=1}^{T_e} \mathbf{z}_t \in [0,1]^{13} \qquad \text{(mean feature vector)} \tag{2}$$

$$\bar{y}_e = \frac{1}{T_e}\sum_{t=1}^{T_e} O_t \in (0,1) \qquad \text{(observed 21CS rate)} \tag{3}$$

The 19,181 individual lines collapse to **48 episode-level observations**. BLFM regresses $\bar{y}_e$ on $\bar{\mathbf{x}}_e$.

Computing the Sufficient Statistic

Suppose Episode 7 has $T_e = 100$ lines:

- 70 lines: Behaviourism=1 $\Rightarrow$ $\bar{x}_{\text{Beh}} = 0.70$
- 20 lines: Cognitivism=1 $\Rightarrow$ $\bar{x}_{\text{Cog}} = 0.20$
- 10 lines: Constructivism=1 $\Rightarrow$ $\bar{x}_{\text{Con}} = 0.10$
- 40 lines: $O_t = 1$ $\Rightarrow$ $\bar{y}_7 = 0.40$

BLFM sees $(\bar{\mathbf{x}}_7, 0.40)$ as a **single data point**. It does not know which specific lines had 21CS active, nor whether the constructivist moves came at the beginning or end.

2.4 Model Specification

2.4.1 Step 1 — Automatic Relevance Determination (ARD) Prior

Instead of assigning all features the same prior variance, BLFM gives each feature f its own *precision* α_f (inverse variance):

$$\alpha_f \sim \text{Gamma}(1,\, 1) \qquad \forall\, f = 1, \ldots, P \tag{4}$$

$$w_f \sim \mathcal{N}\left(0,\, \frac{1}{\alpha_f}\right) \qquad \forall\, f \tag{5}$$

ARD Intuition — Automatic Feature Selection

Think of α_f as a "tightness dial" for feature f.

- **Irrelevant feature**: The data provides no signal about w_f. α_f stays large (the Gamma$(1, 1)$ prior has mode 0, meaning it prefers small but the posterior is driven by likelihood). Large $\alpha_f \Rightarrow$ small variance $1/\alpha_f \Rightarrow w_f \approx 0$. The feature is effectively *switched off*.
- **Relevant feature**: The data pulls α_f toward small values, widening the prior and allowing w_f to be non-zero.

Result: you never need to manually select which of the 13 EP/KD/CP features to include. The model selects them automatically.

Implementation note. The standard deviation of w_f is $1/\sqrt{\alpha_f}$, so in NumPyro: `dist.Normal(0, 1.0/jnp.sqrt(alpha))`. The precision is α_f, not the standard deviation.

2.4.2 Step 2 — Intercept and Random Effects

$$b \sim \mathcal{N}(0,\, 1) \tag{6}$$

$$\sigma_{\text{cell}} \sim \text{Half-Normal}(1) \tag{7}$$

$$u_e \mid \sigma_{\text{cell}} \sim \mathcal{N}(0,\, \sigma^2_{\text{cell}}) \tag{8}$$

u_e is a **per-episode random effect** capturing everything the content features cannot explain: the specific teacher's charisma, the class's prior engagement, time of day, and so on. Episodes sharing the same Subject×Grade cell share a common σ_{cell}, creating **partial pooling**: a Grade-9 Science episode borrows strength from other Grade-9 Science episodes when estimating σ_{cell}.

2.4.3 Step 3 — Predicted Episode Rate

$$\boxed{\mu_e = \sigma\left(\mathbf{w}^\top \bar{\mathbf{x}}_e + u_e + b\right)} \qquad \mu_e \in (0, 1) \tag{9}$$

where $\sigma(x) = 1/(1 + e^{-x})$ is the sigmoid/logistic function that maps any real number to $(0, 1)$.

2.4.4 Step 4 — Beta Regression Likelihood

Since $\bar{y}_e$ is a proportion in $(0, 1)$, a Beta likelihood is the natural choice:

$$\phi \sim \text{Gamma}(5,\ 0.1) \quad \text{(concentration/precision)} \tag{10}$$

$$\bar{y}_e \sim \text{Beta}(\mu_e\phi,\ \ (1-\mu_e)\phi) \tag{11}$$

Beta Distribution Mean and Variance

If $Y \sim \text{Beta}(a, b)$, then:

$$\mathbb{E}[Y] = \frac{a}{a+b}, \qquad \text{Var}[Y] = \frac{\mathbb{E}[Y]\big(1-\mathbb{E}[Y]\big)}{\phi+1}$$

In our parameterisation: $a = \mu_e\phi$, $b = (1-\mu_e)\phi$, so:

$$\mathbb{E}[\bar{y}_e] = \frac{\mu_e\phi}{\mu_e\phi + (1-\mu_e)\phi} = \mu_e$$

The Beta likelihood says: "the observed 21CS rate $\bar{y}_e$ should be close to the predicted rate μ_e, with concentration controlled by ϕ." Larger ϕ = tighter fit. The prior $\phi \sim \text{Gamma}(5, 0.1)$ has mean 50, encoding moderate concentration.

2.5 Full Generative Story

BLFM Generative Process (for all episodes simultaneously)

1. For each feature f: draw precision $\alpha_f \sim \text{Gamma}(1, 1)$
2. For each feature f: draw weight $w_f \sim \mathcal{N}(0, 1/\alpha_f)$
3. Draw global intercept $b \sim \mathcal{N}(0, 1)$
4. Draw cell variance $\sigma_{\text{cell}} \sim \text{Half-Normal}(1)$
5. For each episode e: draw episode effect $u_e \sim \mathcal{N}(0, \sigma^2_{\text{cell}})$
6. For each episode e: compute $\mu_e = \sigma(\mathbf{w}^\top \bar{\mathbf{x}}_e + u_e + b)$
7. Draw concentration $\phi \sim \text{Gamma}(5, 0.1)$
8. For each episode e: **observe** $\bar{y}_e \sim \text{Beta}(\mu_e\phi,\ (1-\mu_e)\phi)$

2.6 Worked Numerical Example

Suppose MCMC returns these posterior means: $w_{\text{Beh}} = -1.5$, $w_{\text{Cog}} = -0.4$, $w_{\text{Con}} = +1.8$, $w_{\text{App}} = +2.1$ (all other $w_f \approx 0$), $b = -0.5$.

Episode A (mostly Behaviourist, low cognitive demand):

$$\bar{x}_{\text{Beh}} = 0.80, \quad \bar{x}_{\text{App}} = 0.05, \quad u_A = +0.1$$

$$\text{logit}(\mu_A) = (-1.5)(0.80) + (2.1)(0.05) - 0.5 + 0.1 = -2.0 \ \Rightarrow\ \hat{\mu}_A = \sigma(-2.0) = \mathbf{0.12}$$

Episode B (more Constructivist, higher Applying):

$$\bar{x}_{\text{Con}} = 0.30, \quad \bar{x}_{\text{App}} = 0.25, \quad u_B = -0.1$$

$$\text{logit}(\mu_B) = (1.8)(0.30) + (2.1)(0.25) - 0.5 - 0.1 = +0.465 \;\Rightarrow\; \hat{\mu}_B = \sigma(0.465) = \mathbf{0.61}$$

The model correctly assigns a much higher predicted 21CS rate to Episode B.

2.7 Code

```
def blfm_model(x_bar, y_bar, cell_idx, n_cells, P):
    """
    x_bar    : shape (E, 13)  -- mean feature vector per
       episode
    y_bar    : shape (E,)     -- observed 21CS rate per episode
    cell_idx : shape (E,)     -- Subject x Grade cell index
    n_cells  : int            -- number of unique cells
    P        : int            -- number of features (13)
    """
# Step 1: ARD priors
    # One precision alpha_f per feature
    alpha = numpyro.sample("alpha",
                           dist.Gamma(1.0, 1.0).expand([P]))
    # Weight: w_f ~ N(0, 1/alpha_f)
    # std = 1/sqrt(alpha) because Var(w_f) = 1/alpha_f
    w = numpyro.sample("w",
                       dist.Normal(jnp.zeros(P),
                                   1.0 / jnp.sqrt(alpha)))
    # Step 2: Global intercept
    b = numpyro.sample("b", dist.Normal(0.0, 1.0))

    # Step 3: Subject x Grade random effects
    # sigma_cell controls how much episodes within a cell vary
    sigma_cell = numpyro.sample("sigma_cell",
                                dist.HalfNormal(1.0))
    # u_cell[c] ~ N(0, sigma_cell^2) for each cell c
    u_cell = numpyro.sample("u_cell",
                 dist.Normal(jnp.zeros(n_cells),
                             sigma_cell * jnp.ones(n_cells)))
    # Map each episode to its cell's random effect
    u_e = u_cell[cell_idx]              # shape: (E,)

    # Step 4: Predicted episode 21CS rate
    # logit(mu_e) = w^T x_bar_e + u_e + b
    logit_mu = jnp.dot(x_bar, w) + u_e + b
    mu = jax.nn.sigmoid(logit_mu)       # mu_e in (0, 1)

    # Step 5: Beta precision
    # Gamma(5, 0.1): mean=50, represents moderate concentration
    phi = numpyro.sample("phi", dist.Gamma(5.0, 0.1))
```

```
40
41      # Step 6: Beta likelihood
42      a  = mu * phi + 1e-6                # alpha shape param
43      b_ = (1.0 - mu) * phi + 1e-6       # beta shape param
44      # Clamp y_bar to open interval: Beta requires (0,1)
            strictly
45      y_bar_clamped = jnp.clip(y_bar, 1e-4, 1.0 - 1e-4)
46      numpyro.sample("y_bar", dist.Beta(a, b_), obs=y_bar_clamped
            )
```

Listing 1: BLFM: Full model definition in NumPyro

2.7.1 Key Implementation Details Explained

`1.0/jnp.sqrt(alpha)`
: The standard deviation of w_f is $\alpha_f^{-1/2}$, because $\mathrm{Var}(w_f) = 1/\alpha_f$. Always distinguish precision (α) from standard deviation.

`u_cell[cell_idx]`
: `cell_idx[i]` stores which Subject×Grade cell episode i belongs to. This single indexing operation maps each of the E episodes to its cell's random effect.

`jnp.clip(y_bar, 1e-4, 1-1e-4)`
: The Beta distribution requires strict open-interval support. Any episode with $\bar{y}_e = 0$ (no 21CS at all) or $\bar{y}_e = 1$ would produce $\log \mathrm{Beta}(-\infty)$. The clamp prevents this.

`1e-6` **stabiliser on** a **and** $b_$
: When $\mu \approx 0$ or $\mu \approx 1$, one of the Beta shape parameters would be nearly zero, causing numerical underflow. The small additive constant stabilises the computation.

2.8 Running MCMC

```
1 kernel_blfm = NUTS(blfm_model, target_accept_prob=0.90)
2 mcmc_blfm   = MCMC(kernel_blfm,
3                    num_warmup=500,     # adapt step size for 500
                         steps
4                    num_samples=1000,   # draw 1000 posterior
                         samples
5                    num_chains=2,       # run 2 independent
                         chains
6                    progress_bar=True)
7 mcmc_blfm.run(jax.random.PRNGKey(42),
8               x_bar_jax, y_bar_jax,
9               cell_idx_jax, n_cells, P)
```

Listing 2: BLFM: NUTS sampler configuration

NUTS (No-U-Turn Sampler) automatically tunes step size and trajectory length during warmup, making it far more efficient than basic Metropolis-Hastings for high-dimensional

posteriors. `target_accept_prob=0.90` means: aim for 90% acceptance, which NUTS achieves by tuning step sizes during warmup.

2.9 What BLFM Cannot Answer

BLFM Blind Spots

- What is the probability the classroom is productive *right now*, at line 47?
- Did the Constructivist move at line 32 *cause* a regime shift?
- How does the productive state evolve *across* the lesson arc?

These require a sequential model with a latent state. Enter PST.

3 Model 2: PST — Probabilistic Skill Tracing (Sequential BKT)

3.1 Core Question

PST Answers

How often does the classroom switch between productive and unproductive discourse? And what is the probability of being in the productive regime at each line, without knowing anything about the pedagogical content?

3.2 Hidden Markov Model (HMM) Primer

PST models the classroom as a two-state Hidden Markov Model. We have:

- A **latent state** $M_t \in \{0, 1\}$ at each line t — never directly observed.
- A **visible observation** $O_t \in \{0, 1\}$ — whether 21CS is active.

The HMM diagram below shows the structure:

3.3 PST Parameters

Symbol	Meaning	Prior
π_0	$P(M_1 = 1)$: probability episode starts in the productive state	Beta$(2, 3)$
p_L	Learning rate: $P(M_t{=}1 \mid M_{t-1}{=}0)$ per line	Beta$(1.5, 20)$
p_F	Forgetting rate: $P(M_t{=}0 \mid M_{t-1}{=}1)$ per line	Beta$(1, 50)$
p_S	Slip: $P(O_t{=}0 \mid M_t{=}1)$	**Fixed** $= 0.015$
p_G	Guess: $P(O_t{=}1 \mid M_t{=}0)$	**Fixed** $= 0.009$

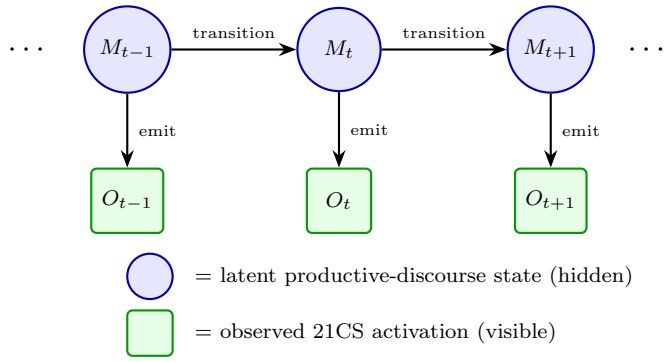


Figure 2: PST graphical model. The discourse state M_t transitions between productive (M=1) and unproductive (M=0) with fixed probabilities p_L (learning) and p_F (forgetting), independent of pedagogical content. The observed 21CS activation O_t is emitted with slip probability p_S and guess probability p_G. No covariate vector $\mathbf{z}_t$ is present: content-blindness is intentional, isolating pure regime dynamics from pedagogical content. Compare IO-BKT (Fig. 3), which adds an orange $\mathbf{z}_t$ row and content-driven transition probabilities.

3.3.1 The Transition Matrix

At every time step, the probability of moving between states is constant (PST ignores content):

$$\mathbf{A} = \begin{pmatrix} P(M_t{=}0 \mid M_{t-1}{=}0) & P(M_t{=}1 \mid M_{t-1}{=}0) \\ P(M_t{=}0 \mid M_{t-1}{=}1) & P(M_t{=}1 \mid M_{t-1}{=}1) \end{pmatrix} = \begin{pmatrix} 1-p_L & p_L \\ p_F & 1-p_F \end{pmatrix} \tag{12}$$

The posterior mean $\hat{p}_L = 0.073$ means each line has only a 7.3% chance of transitioning from unproductive to productive. Regimes are sticky.

3.3.2 Emission Probabilities

$$P(O_t = 1 \mid M_t = 1) = 1 - p_S = 0.985 \quad \text{(productive} \Rightarrow \text{21CS almost always observed)} \tag{13}$$

$$P(O_t = 0 \mid M_t = 1) = p_S = 0.015 \quad \text{(“slip”: productive but no 21CS)} \tag{14}$$

$$P(O_t = 1 \mid M_t = 0) = p_G = 0.009 \quad \text{(“guess”: unproductive but 21CS fires)} \tag{15}$$

$$P(O_t = 0 \mid M_t = 0) = 1 - p_G = 0.991 \quad \text{(unproductive} \Rightarrow \text{no 21CS almost always)} \tag{16}$$

Why Call Them “Slip” and “Guess”?

These terms originate from BKT in tutoring systems (Corbett & Anderson, 1994):

- **Slip** ($p_S = 0.015$): Even when the classroom is genuinely in the productive regime, expert raters occasionally miss 21CS activation (momentary pause, annotation noise). This is rare: 21CS is almost always visible when the regime is truly productive.
- **Guess** ($p_G = 0.009$): Even in an unproductive regime, a single spontaneous student question might briefly activate 21CS. Also very rare.

Low slip and guess values mean the states are *nearly fully separable* by their emissions.

3.3.3 Why Are p_S and p_G Fixed?

This is a critical design decision. If we estimated p_S and p_G freely, we would face **label switching**: mathematically, swapping the labels "productive" and "unproductive" (and simultaneously swapping emission parameters) gives the exact same likelihood. MCMC would explore both interpretations, producing posterior means of $p_S \approx p_G \approx 0.5$ — meaningless.

Label Switching Visualised

Imagine the model has two possible solutions:

	Solution A (correct)	**Solution B (flipped)**
State 1 meaning	Productive	Unproductive
p_S (State 1 slip)	0.015	0.991
p_G (State 0 guess)	0.009	0.985
Likelihood	$\mathcal{L}$	$\mathcal{L}$ (identical!)

Both solutions have the same likelihood. Without content features to distinguish states, MCMC mixes freely between them. Fixing p_S and p_G at values from IO-BKT (which can identify them via content features) resolves the ambiguity.

3.4 The Forward Algorithm: Step-by-Step Derivation

The forward algorithm is the engine of PST. It computes the marginal likelihood $P(O_1, \ldots, O_{T_e} \mid \theta)$ *exactly* in $O(T_e)$ time, by recursively integrating out the hidden states M_t.

3.4.1 Why Not Enumerate All State Sequences?

A brute-force approach would sum over all 2^{T_e} state sequences. For $T_e = 384$, this is $\sim 10^{115}$ terms — impossible. The forward algorithm avoids this by exploiting the **Markov property**: M_t depends only on M_{t-1}, not the full history.

3.4.2 Notation

Define the **filtered joint probability** at time t:

$$\alpha_t^{(m)} \;=\; P(M_t = m,\; O_1, O_2, \ldots, O_t \mid \theta) \tag{17}$$

This is the probability that: (a) we have seen observations $O_1, \ldots, O_t$, *and* (b) the state at time t is m.

3.4.3 Initialisation ($t = 1$)

$$\alpha_1^{(0)} = P(M_1 = 0) \cdot P(O_1 \mid M_1 = 0) = (1 - \pi_0) \cdot e_0(O_1) \tag{18}$$

$$\alpha_1^{(1)} = P(M_1 = 1) \cdot P(O_1 \mid M_1 = 1) = \pi_0 \cdot e_1(O_1) \tag{19}$$

where the emission shorthand is:

$$e_m(O_t) = \begin{cases} 1 - p_G & \text{if } m = 0,\ O_t = 0 \\ p_G & \text{if } m = 0,\ O_t = 1 \\ p_S & \text{if } m = 1,\ O_t = 0 \\ 1 - p_S & \text{if } m = 1,\ O_t = 1 \end{cases} \tag{20}$$

Normalise: $c_1 = \alpha_1^{(0)} + \alpha_1^{(1)} + \varepsilon$; then $\alpha_1^{(m)} \leftarrow \alpha_1^{(m)} / c_1$.

3.4.4 Recursion ($t \geq 2$): Predict–Update

Predict step — marginalise over the previous state:

$$\tilde{\alpha}_t^{(0)} = \alpha_{t-1}^{(0)} (1 - p_L) + \alpha_{t-1}^{(1)} p_F \tag{21}$$

$$\tilde{\alpha}_t^{(1)} = \alpha_{t-1}^{(0)} p_L + \alpha_{t-1}^{(1)} (1 - p_F) \tag{22}$$

Update step — weight by the emission probability:

$$\alpha_t^{(0)} \propto \tilde{\alpha}_t^{(0)} \cdot e_0(O_t) \tag{23}$$

$$\alpha_t^{(1)} \propto \tilde{\alpha}_t^{(1)} \cdot e_1(O_t) \tag{24}$$

Normalise with $c_t = \alpha_t^{(0)} + \alpha_t^{(1)} + \varepsilon$.

3.4.5 Log-Likelihood

$$\log P(O_1, \ldots, O_{T_e} \mid \theta) \ = \ \sum_{t=1}^{T_e} \log c_t \tag{25}$$

This is the quantity NUTS maximises (in expectation) during MCMC.

3.5 Worked Numerical Forward Pass

Parameters: $\pi_0 = 0.4$, $p_L = 0.07$, $p_F = 0.02$, $p_S = 0.015$, $p_G = 0.009$.

Observations: $[O_1, O_2, O_3, O_4, O_5] = [1, 1, 0, 1, 1]$.

Emission probabilities: $e_0(O = 1) = 0.009$, $e_0(O = 0) = 0.991$, $e_1(O = 1) = 0.985$, $e_1(O = 0) = 0.015$.

t	O_t	$\tilde{\alpha}^{(0)}$	$\tilde{\alpha}^{(1)}$	$\alpha^{(0)}$ (normalised)	$\alpha^{(1)}$ (normalised)	c_t
1	1	—	—	0.0135	**0.9865**	0.3994
2	1	0.0323	0.9677	0.0003	**0.9997**	0.9535
3	0	0.0203	0.9797	**0.5777**	0.4223	0.0348
4	1	0.5457	0.4543	0.0109	**0.9891**	0.4524
5	1	0.0299	0.9701	0.0003	**0.9997**	0.9559

Reading the Forward Table

- $t = 1$: We observe $O_1 = 1$. Since $e_1(1) = 0.985 \gg e_0(1) = 0.009$, the 21CS observation strongly implies $M_1 = 1$: $\hat{P}(M_1 = 1) = 0.9865$.
- $t = 2$: Observing $O_2 = 1$ again. The predicted state before updating: $\tilde{\alpha}^{(0)} = (0.0135)(0.93) + (0.9865)(0.02) = 0.0323$. After the update with $O = 1$: $P(M_2 = 1)$ rises even higher to 0.9997.
- $t = 3$: $O_3 = 0$! No 21CS observed. The update weight for $M = 1$ is now $e_1(0) = p_S = 0.015$ (tiny), so $P(M_3 = 1)$ collapses to 0.4223. The classroom could have slipped out of the productive regime.
- $t = 4$: $O_4 = 1$ again. The state immediately recovers to $P(M_4 = 1) = 0.9891$. This is the near-permanent regime in action: one negative observation cannot permanently kill a productive state when p_S is small.

3.6 Prior Choices Explained

Prior	Mean	Reasoning
$\pi_0 \sim \text{Beta}(2, 3)$	0.40	Slight prior that episodes start unproductively. Most Indian classroom episodes (Behaviourist-dominant) begin with routine instruction.
$p_L \sim \text{Beta}(1.5, 20)$	0.07	Small learning rate expected. Consistent with near-permanent regime hypothesis: productive states are rare achievements, not the default.
$p_F \sim \text{Beta}(1, 50)$	0.02	Forgetting even rarer than learning. Once a productive regime is established, it tends to persist.

3.7 Key Result: Regime Persistence

$$\hat{p}_L = 0.073 \ \ [0.067,\ 0.080], \qquad \hat{p}_F = 0.021 \ \ [0.012,\ 0.033] \tag{26}$$

Average lines per transition: $1/\hat{p}_L \approx 14$. A 384-line episode sees roughly $384 \times 0.073 \approx 28$ transitions total — but most are micro-fluctuations, not sustained regime changes.

3.8 Code

```
EPS      = 1e-7
PS_FIXED = 0.015    # from IO-BKT posterior (fixes label
    switching)
PG_FIXED = 0.009    # from IO-BKT posterior

def pst_forward_single(obs_e, mask_e, pi0, p_L, p_F):
    """
```

```
    obs_e  : shape (Tmax,) -- 21CS observations (0/1), padded
    mask_e : shape (Tmax,) -- 1 for real lines, 0 for padding
    pi0    : scalar        -- P(M_1=1)
    p_L    : scalar        -- learning rate (0->1 per line)
    p_F    : scalar        -- forgetting rate (1->0 per line)
    """
    pS = PS_FIXED
    pG = PG_FIXED

    # Initialisation at t=0
    o0 = obs_e[0]
    # Emission: P(O_1 | M_1=0) and P(O_1 | M_1=1)
    e0_0 = jnp.where(o0 == 1., pG,      1. - pG)
    e1_0 = jnp.where(o0 == 1., 1. - pS, pS)
    # alpha_1^(0) and alpha_1^(1), unnormalised
    a0 = (1. - pi0) * e0_0
    a1 = pi0        * e1_0
    # Normalise and log the normalisation constant
    c0 = a0 + a1 + EPS
    a0 = a0 / c0;  a1 = a1 / c0
    ll0 = jnp.log(c0) * mask_e[0]  # 0 if padding line

    # Recursion: t=1, 2, ..., Tmax-1
    def step(carry, t):
        a0, a1, ll = carry
        o_t = obs_e[t]
        m_t = mask_e[t]   # padding guard

        # PREDICT: marginalise over M_{t-1}
        pred0 = a0 * (1. - p_L) + a1 * p_F        # Eq.(12)
        pred1 = a0 * p_L        + a1 * (1. - p_F)  # Eq.(13)

        # Emission probabilities for current observation
        e0 = jnp.where(o_t == 1., pG,      1. - pG)
        e1 = jnp.where(o_t == 1., 1. - pS, pS)

        # UPDATE: weight by emission
        b0 = pred0 * e0
        b1 = pred1 * e1
        c  = b0 + b1 + EPS

        # Accumulate log-likelihood (zero for padding)
        return (b0/c, b1/c, ll + jnp.log(c) * m_t), None

    # lax.scan: JIT-compiled loop, runs efficiently on GPU
    (_, _, ll_final), _ = lax.scan(
        step, (a0, a1, ll0), jnp.arange(1, Tmax)
    )
    return ll_final
```

Listing 3: PST: Forward algorithm for one episode

```python
# Vectorise forward pass: apply to all E episodes at once
pst_forward_all = vmap(pst_forward_single,
                       in_axes=(0, 0, None, None, None))

def pst_model(obs, mask):
    """
    Only 3 parameters are estimated -- content-free by design.
    """
    # Beta(2,3): mean=0.40, slight prior for unproductive start
    pi0 = numpyro.sample("pi0", dist.Beta(2.0,  3.0))
    # Beta(1.5,20): mean=0.07, small learning rate expected
    p_L = numpyro.sample("p_L", dist.Beta(1.5, 20.0))
    # Beta(1,50): mean=0.02, forgetting even rarer
    p_F = numpyro.sample("p_F", dist.Beta(1.0, 50.0))

    # Forward pass on all episodes simultaneously
    ll = pst_forward_all(obs, mask, pi0, p_L, p_F)
    # numpyro.factor adds log-prob directly to the model
    numpyro.factor("seq_loglik", jnp.sum(ll))

# Run NUTS MCMC
kernel_pst = NUTS(pst_model, target_accept_prob=0.90)
mcmc_pst   = MCMC(kernel_pst, num_warmup=500,
                  num_samples=1000, num_chains=2)
mcmc_pst.run(jax.random.PRNGKey(42), obs_jax, mask_jax)
```

Listing 4: PST: Model and MCMC

3.8.1 Two Technical Details

`jax.lax.scan` A standard Python `for` loop would unroll to $T_{\max}$ Python-level operations, each launching a separate GPU kernel — very slow. `lax.scan` compiles the entire loop into a single XLA kernel, running the forward pass on GPU in milliseconds.

The `mask` trick All episodes are zero-padded to a uniform length $T_{\max}$. The mask ensures padding lines contribute $\log(c_t) \times 0 = 0$ to the log-likelihood. Without masking, the algorithm would treat padding zeros as real "no-21CS" observations, artificially inflating the estimated p_F.

3.9 Why PST's AUC is Misleading

PST achieves AUC = 0.928, higher than IO-BKT's 0.873. But this is not a genuine win.

PST AUC is Autocorrelation, Not Insight

PST's one-step-ahead prediction of O_t is:

$$\hat{P}(O_t = 1 \mid O_1, \ldots, O_{t-1}) \approx O_{t-1}$$

The correlation $\text{corr}(P_t, O_{t-1}) = 0.998$ confirms this. PST is essentially saying "the next observation looks like the current one," which is a tautology when regimes barely change. There is no pedagogical insight: IO-BKT's 0.873, achieved with *content features alone* (not the observation history), is the honest comparison.

4 Model 3: IO-BKT — Input-Output Bayesian Knowledge Tracing

4.1 Core Question

IO-BKT Answers

Which EP/KD/CP codes are most strongly associated with transitions into and out of the productive discourse regime at each utterance? The transition coefficients $\boldsymbol{\beta}_{\text{on}}$ and $\boldsymbol{\beta}_{\text{off}}$ quantify these associations from observational data; causal interpretation requires replication with experimental or quasi-experimental designs.

4.2 The Key Innovation: Content-Driven Transitions

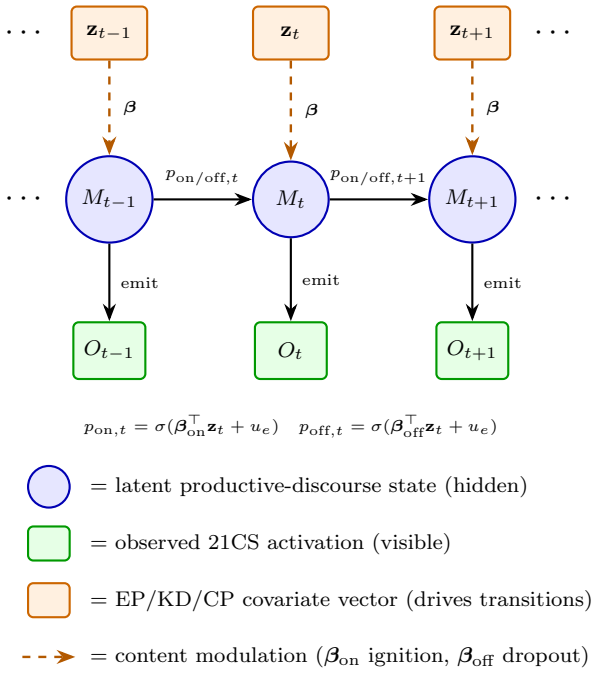


Figure 3: IO-BKT graphical model. At each line t, the observed EP/KD/CP covariate vector $\mathbf{z}_t$ modulates both the 0→1 transition probability (ignition, via $\boldsymbol{\beta}_{\text{on}}$) and the 1→0 transition probability (dropout, via $\boldsymbol{\beta}_{\text{off}}$), making transitions content-driven rather than fixed. The latent productive-discourse state M_t is inferred via the forward algorithm; 21CS activation O_t is the observed emission. Compare PST (Fig. 2), which has no $\mathbf{z}_t$ row and uses constant transition probabilities.

PST uses a constant transition probability: every line has the same p_L (chance of $0 \to 1$)

regardless of whether the teacher is asking "What is the formula?" or "Now apply this to design a bridge." This is clearly wrong.

IO-BKT replaces constant transitions with **sigmoid-linear functions of $\mathbf{z}_t$**:

$$\boxed{P(M_t = 1 \mid M_{t-1} = 0,\ \mathbf{z}_t) = \sigma\big(\boldsymbol{\beta}_{\text{on}}^\top \mathbf{z}_t + u_e\big)} \tag{27}$$

$$\boxed{P(M_t = 0 \mid M_{t-1} = 1,\ \mathbf{z}_t) = \sigma\big(\boldsymbol{\beta}_{\text{off}}^\top \mathbf{z}_t + u_e\big)} \tag{28}$$

The Two Coefficient Vectors Have Distinct Roles

- $\boldsymbol{\beta}_{\text{on}}$ (**ignition coefficients**): $\beta_{\text{on},f} > 0$ means feature f *promotes* entry into the productive state from unproductive. $\beta_{\text{on},f} < 0$ means feature f *suppresses* ignition.
- $\boldsymbol{\beta}_{\text{off}}$ (**dropout coefficients**): $\beta_{\text{off},f} > 0$ means feature f *promotes dropout* from productive to unproductive. $\beta_{\text{off},f} < 0$ means feature f *prevents dropout* (sustains the productive state).

A feature can have *opposite* signs in $\boldsymbol{\beta}_{\text{on}}$ and $\boldsymbol{\beta}_{\text{off}}$ — for example, Applying has $\beta_{\text{on}} = +3.41$ (ignites) *and* $\beta_{\text{off}} = -3.40$ (sustains). A simple regression or correlation cannot separate these roles.

4.3 Full Model Specification

4.3.1 Priors

$$\boldsymbol{\beta}_{\text{on}},\ \boldsymbol{\beta}_{\text{off}} \sim \mathcal{N}(\mathbf{0},\ \mathbf{I}_{14}) \tag{29}$$

$$\pi_0 \sim \text{Beta}(2,\ 3) \tag{30}$$

$$p_S \sim \text{Beta}(2,\ 30)\ \ [\text{mean} \approx 0.06] \tag{31}$$

$$p_G \sim \text{Beta}(1.5,\ 30)\ \ [\text{mean} \approx 0.05] \tag{32}$$

$$\sigma_{\text{cell}} \sim \text{Half-Normal}(1) \tag{33}$$

$$u_e \sim \mathcal{N}(0,\ \sigma_{\text{cell}}^2) \tag{34}$$

Unlike PST, IO-BKT *estimates* p_S and p_G. The content features break label-switching symmetry: by the time MCMC runs, it knows Applying (+3.41) is associated with ignition, so the "productive" label attaches to the state where high-CP lines drive entry — there is no ambiguity.

4.3.2 Emission Model

Unchanged from PST (Eqs. 13-16), but now p_S and p_G are *estimated*:

$$P(O_t = 1 \mid M_t = 1) = 1 - p_S, \qquad P(O_t = 1 \mid M_t = 0) = p_G \tag{35}$$

The posterior means are $\hat{p}_S = 0.015$, $\hat{p}_G = 0.009$ — confirming PST's fixed values were appropriate.

4.3.3 Forward Algorithm with Time-Varying Transitions

The forward algorithm is identical in structure to PST, but now p_L and p_F are replaced by line-specific values $p_{\text{on},t}$ and $p_{\text{off},t}$:

Predict step:

$$\tilde{\alpha}_t^{(0)} = \alpha_{t-1}^{(0)}\,(1 - p_{\text{on},t}) + \alpha_{t-1}^{(1)}\,p_{\text{off},t} \tag{36}$$

$$\tilde{\alpha}_t^{(1)} = \alpha_{t-1}^{(0)}\,p_{\text{on},t} + \alpha_{t-1}^{(1)}\,(1 - p_{\text{off},t}) \tag{37}$$

Update step: same as PST (weight by emission). The key difference: every line has its *own* $p_{\text{on},t}$ and $p_{\text{off},t}$, computed from $\mathbf{z}_t$ via sigmoid.

4.4 Worked Numerical Example: Transition Probabilities

Using the paper's posterior means ($\boldsymbol{\beta}_{\text{on}}$ and $\boldsymbol{\beta}_{\text{off}}$ from Table II, intercept $u_e = 0$):

Line A — Behaviourism + Remembering: $\mathbf{z}_{\text{A}} = [\underbrace{1,0,0}_{\text{EP}},\underbrace{0,0,0,0}_{\text{KD}},\underbrace{1,0,0,0,0,0}_{\text{CP}},1]$

$$\boldsymbol{\beta}_{\text{on}}^{\top}\mathbf{z}_{\text{A}} = (-1.50)_{\text{Beh}} + (-1.15)_{\text{Rem}} + (-1.78)_{\text{int}} = -4.43$$

$$p_{\text{on,A}} = \sigma(-4.43) = \mathbf{0.012} \;\; (1.2\%\text{ ignition chance})$$

Line B — Constructivism + Applying: $\mathbf{z}_{\text{B}} = [\underbrace{0,0,1}_{\text{EP}},\underbrace{0,0,0,0}_{\text{KD}},\underbrace{0,0,1,0,0,0}_{\text{CP}},1]$

$$\boldsymbol{\beta}_{\text{on}}^{\top}\mathbf{z}_{\text{B}} = (+1.25)_{\text{Con}} + (+3.41)_{\text{App}} + (-1.78)_{\text{int}} = +2.88$$

$$p_{\text{on,B}} = \sigma(+2.88) = \mathbf{0.947} \;\; (94.7\%\text{ ignition chance})$$

What This Means for a Teacher

If a teacher is giving a Behaviourist recall drill (Line A type), each line has only a 1.2% chance of igniting a productive discourse regime. But if the teacher shifts to a Constructivist application task (Line B type), the probability jumps to 94.7%. The IO-BKT coefficients operationalise precisely *which* moves matter and by how much.

4.5 Interpreting Table II: The Posterior Coefficients

Ignition ($\boldsymbol{\beta}_{\text{on}}$)			Dropout prevention / promotion ($\boldsymbol{\beta}_{\text{off}}$)		
Feature	**Coeff.**	$\lvert\mu\rvert > 2\text{SD}$	**Feature**	**Coeff.**	$\lvert\mu\rvert > 2\text{SD}$
Applying	+2.34	⋆	Applying	−3.42	⋆
Constructivism	+1.60		Constructivism	−2.70	⋆
Behaviourism	−0.93	⋆	Cognitivism	−1.18	⋆
Remembering	−2.06	⋆	Remembering	+0.48	⋆
Procedural	−0.15	⋆	Understanding	−0.29	⋆
Intercept	−2.88	⋆	Intercept	−0.06	⋆

Reading each coefficient:

Applying, $\beta_{\text{on}} = +2.34^{\star}$ and $\beta_{\text{off}} = -3.42^{\star}$

The dominant lever in both directions. Applying-level cognitive demand (Bloom's level 3: "use knowledge in a new situation") both *ignites* the productive state and *sustains* it once established. It is the most impactful feature in the model.

Constructivism, $\beta_{\text{off}} = -2.70^{\star}$

Once the classroom is productive, Constructivist moves (Vygotsky zone-of-proximal-development scaffolding, collaborative discovery) are the strongest *prevention of dropout*. Note: $\beta_{\text{on}} = +1.60$ for Constructivism is *not* credibly non-zero (< 2SD), meaning Constructivism sustains but does not reliably *ignite* (likely due to its extreme rarity in the corpus: $< 5\%$ of lines).

Behaviourism, $\beta_{\text{on}} = -0.93^{\star}$

Operant-conditioning-style instruction (drill, rote recall, reward/punish) actively *suppresses* entry into the productive regime.

Remembering, $\beta_{\text{on}} = -2.06^{\star}$ and $\beta_{\text{off}} = +0.48^{\star}$

Pure recall tasks both suppress ignition *and* are the most reliable dropout trigger. If the classroom is productive and the teacher reverts to "remember this definition," the productive state is at serious risk.

Intercept, $\beta_{\text{on}} = -2.888^{\star}$

The baseline ignition probability (when all features are zero) is $\sigma(-2.88) = 0.$.

Classrooms default toward unproductive discourse: productivity requires active effort.

4.6 Code

```
def forward_single(obs_e, mask_e, pon_e, poff_e, pi0, pS, pG):
    """
    pon_e  : shape (Tmax,) -- P(M_t=1|M_{t-1}=0, z_t), per line
    poff_e : shape (Tmax,) -- P(M_t=0|M_{t-1}=1, z_t), per line
    These vary each line -- the key difference from PST!
    """
    # Initialise
    o0   = obs_e[0]
    e0_0 = jnp.where(o0==1., pG, 1.-pG)      # P(O1|M1=0)
    e1_0 = jnp.where(o0==1., 1.-pS, pS)      # P(O1|M1=1)
    a0   = (1.-pi0) * e0_0
    a1   = pi0      * e1_0
    c0   = a0 + a1 + EPS
    a0  /= c0;  a1 /= c0
    ll0  = jnp.log(c0) * mask_e[0]

    def step(carry, t):
        a0, a1, ll = carry
        o_t    = obs_e[t]
```

```
        m_t    = mask_e[t]
        pon_t  = pon_e[t]    # content-driven -- changes each
            line
        poff_t = poff_e[t]   # content-driven -- changes each
            line

        # PREDICT: Eqs. (16)-(17)
        pred0 = a0*(1.-pon_t) + a1*poff_t
        pred1 = a0*pon_t      + a1*(1.-poff_t)

        # UPDATE: weight by emission
        e0 = jnp.where(o_t==1., pG, 1.-pG)
        e1 = jnp.where(o_t==1., 1.-pS, pS)
        b0 = pred0*e0;  b1 = pred1*e1
        c  = b0 + b1 + EPS

        return (b0/c, b1/c, ll + jnp.log(c)*m_t), None

    (_, _, ll_final), _ = lax.scan(
        step, (a0, a1, ll0), jnp.arange(1, Tmax)
    )
    return ll_final

# Vectorise: run forward_single on all E episodes in parallel
forward_all = vmap(forward_single,
                   in_axes=(0, 0, 0, 0, None, None, None))
```

Listing 5: IO-BKT: Forward algorithm with time-varying transitions

```
def iobkt_model(Z, obs, mask, cell_idx, n_cells, Tmax, P):
    """
    Z       : shape (E, Tmax, P) -- feature vectors, all
       episodes
    obs     : shape (E, Tmax)    -- 21CS observations
    mask    : shape (E, Tmax)    -- padding mask
    """
    E = Z.shape[0]

    # Global coefficient vectors (14D)
    # b_on[f]:  how feature f drives productive ignition
    # b_off[f]: how feature f drives productive dropout
    b_on  = numpyro.sample("b_on",
                dist.Normal(jnp.zeros(P), jnp.ones(P)))
    b_off = numpyro.sample("b_off",
                dist.Normal(jnp.zeros(P), jnp.ones(P)))

    #  Subject x Grade random effects
    sigma_cell = numpyro.sample("sigma_cell",
                                dist.HalfNormal(1.0))
    u_cell = numpyro.sample("u_cell",
```

```
                    dist.Normal(jnp.zeros(n_cells),
                                sigma_cell * jnp.ones(n_cells)))

    # HMM parameters
    pi0 = numpyro.sample("pi0", dist.Beta(2.0,  3.0))
    pS  = numpyro.sample("pS",  dist.Beta(2.0, 30.0)) # mean ~0.06
    pG  = numpyro.sample("pG",  dist.Beta(1.5, 30.0)) # mean ~0.05

    # Compute time-varying transition probabilities
    u_e = u_cell[cell_idx]              # shape: (E,)
    # eta_on[e,t]  = sum_p (Z[e,t,p] * b_on[p]) + u_e[e]
    # "etp,p->et" means: sum over last axis (features p)
    eta_on  = jnp.einsum("etp,p->et", Z, b_on)  + u_e[:, None]
    eta_off = jnp.einsum("etp,p->et", Z, b_off) + u_e[:, None]
    # Squeeze through sigmoid to get probabilities
    p_on  = jax.nn.sigmoid(eta_on)   # shape: (E, Tmax)
    p_off = jax.nn.sigmoid(eta_off)  # shape: (E, Tmax)
    # p_on[e,t]  = P(M_t=1 | M_{t-1}=0, z_{e,t})
    # p_off[e,t] = P(M_t=0 | M_{t-1}=1, z_{e,t})

    # Run forward algorithm over all episodes
    ll_per_episode = forward_all(obs, mask, p_on, p_off,
                                 pi0, pS, pG)
    numpyro.factor("seq_loglik", jnp.sum(ll_per_episode))
```

Listing 6: IO-BKT: NumPyro model with hierarchical effects

4.6.1 The `einsum` Operation Decoded

```
eta_on = jnp.einsum("etp,p->et", Z, b_on) + u_e[:, None]
```

This computes for all episodes e and time steps t simultaneously:

$$\eta_{\text{on},e,t} = \sum_{p=1}^{P} Z_{e,t,p} \cdot \beta_{\text{on},p} + u_e \;=\; \boldsymbol{\beta}_{\text{on}}^{\top} \mathbf{z}_{e,t} + u_e$$

The index string `"etp,p->et"` means: *sum over dimension p (features), keeping dimensions e (episodes) and t (time steps).* This replaces a nested Python loop with a single GPU matrix operation. Without `einsum`, you would write:

```
# Equivalent but slow:
for e in range(E):
    for t in range(Tmax):
        eta_on[e,t] = sum(Z[e,t,:] * b_on) + u_e[e]
```

4.6.2 MCMC Configuration

```
kernel = NUTS(
    iobkt_model,
    target_accept_prob=0.95,    # higher than BLFM's 0.90
    max_tree_depth=10,          # allow longer HMC trajectories
)
mcmc = MCMC(
    kernel,
    num_warmup=500,
    num_samples=1000,
    num_chains=2,
    chain_method='sequential', # avoid GPU memory doubling
    progress_bar=True,
)
```

Listing 7: IO-BKT: NUTS with tighter tolerances for complex geometry

Why `target_accept_prob=0.95`? IO-BKT's posterior has a more complex, curved geometry than BLFM's, because the 28 coefficients ($\boldsymbol{\beta}_{\text{on}}$ and $\boldsymbol{\beta}_{\text{off}}$) interact with the forward algorithm non-linearly. A higher target acceptance forces NUTS to take smaller, more careful steps (via a smaller step size), reducing the number of divergent transitions at the cost of slower exploration. This is the standard recommendation when you see many divergences with the default 0.80 or 0.90.

4.7 Label Switching: Detection and Correction

```
pS_mean = float(np.array(samples["pS"]).mean())
if pS_mean > 0.5:
    # The model learned the states in REVERSE order.
    # "State 1" actually behaves like the unproductive state.
    print("Label switching detected -- flipping states")

    # Flip all coefficient signs: if b_on drove 0->1 transition
    # in the wrong direction, negating it flips the roles.
    samples["b_on"]  = -samples["b_on"]
    samples["b_off"] = -samples["b_off"]

    # Correct emission parameters.
    # Original "state 1" had slip-like pS = 1 - pG_correct.
    # After flipping:  new pS = 1 - old pG,  new pG = 1 - old
       pS
    pS_old = samples["pS"].copy()
    pG_old = samples["pG"].copy()
    samples["pS"] = 1.0 - pG_old
    samples["pG"] = 1.0 - pS_old
```

Listing 8: Label switching check and correction after MCMC

Why does this happen? If MCMC's random initialisation starts near the "wrong" mode (where state 0 is actually more productive than state 1), it may converge to the label-swapped solution. The diagnostic: $\hat{p}_S > 0.5$ would mean the "productive" state has

a 50%+ chance of producing no 21CS — absurd. Negating the coefficients and swapping emission parameters corrects this.

4.8 The Full-Length Normalisation (Tmax Extension)

The training used $T_{\max} = 600$ (92nd percentile). Some episodes (e.g. Episode 37 with 921 lines) were truncated. The normalisation extension re-runs *only the forward pass* with $T_{\max} = 950$, keeping all MCMC-estimated parameters unchanged.

```
# b_on_m, b_off_m etc. are the posterior MEANS (fixed, from MCMC)
# We just extend the input arrays and re-apply the forward pass.

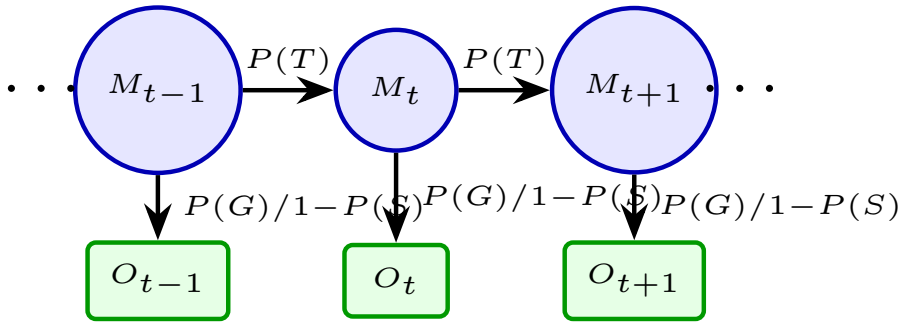
for i in range(E):
    T_new = int(mask_full[i].sum())   # now using all lines
    u_e   = float(u_cell_m[cell_idx[i]])

    # Time-varying transitions for ALL T_new lines
    # (same formula as training, just more timesteps)
    pon_e  = sigmoid(Z_full[i, :T_new] @ b_on_m  + u_e)
    poff_e = sigmoid(Z_full[i, :T_new] @ b_off_m + u_e)

    # Run the same forward algorithm over extended sequence
    probs = forward_pass(obs_full[i, :T_new], pon_e, poff_e,
        ...)
```

Listing 9: Forward pass extension without re-running MCMC

This is valid because the IO-BKT coefficients describe how EP/KD/CP features drive transitions *in general*, they are not specific to the first 600 lines. Applying them to lines 601-950 is legitimate extrapolation, not re-fitting.



one global $P(L_0), P(T), P(G), P(S)$, shared by all 48 episodes;
$M_{t-1}{=}1 \Rightarrow M_t{=}1$ (no forgetting) — no $\mathbf{z}_t$, no per-episode term

Figure 4: BKT aggregate: a single, content-blind, corpus-pooled skill ("discourse").

5 Experimental Setup

Metrics. Following Scarlatos et al. [1], three metrics have been used namely Accuracy (Acc), AUC, and F1 at line level (n=18,506; first line of each episode excluded as no prior-observation context available). AUC is used as the primary metric. Predictions are one-step-ahead: $P(O_t{=}1 \mid O_{1:t-1})$ computed before O_t is revealed. Verified by lag-correlation test: $\text{corr}(P_t, O_{t-1})$=0.998, $\text{corr}(P_t, O_t)$=0.766 for PST with no leakage.

Evaluation: Bayesian models use posterior predictive distributions (regularising priors mitigate overfitting). DKT uses LOO-CV. BKT uses in-sample forward smoother.

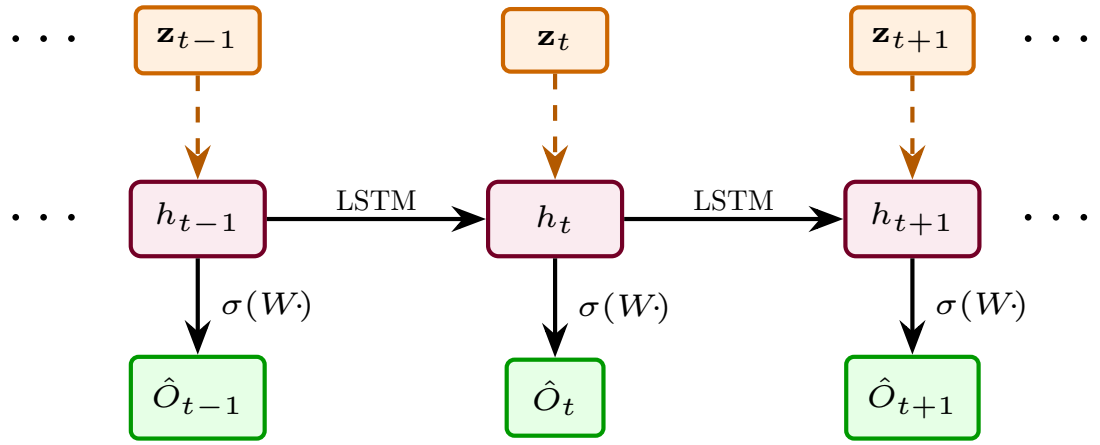
6 Results and Discussions

6.1 Baselines

The three baselines span the design space and each serves a specific evidentiary role; they were not selected for convenience but to answer three distinct questions about what the proposed framework adds over existing tools.

Frequentist BKT (pyBKT) is the field's default sequential KT method. We run BKT aggregate (one KC across all lines), each with 5 random initialisations. Its role is diagnostic: if standard BKT works, the problem is already solved. That this variants collapse to AUC = 0.500 is itself a result – it establishes that near-permanent regimes are precisely the failure mode classical KT is not designed for, because the filtered posterior assigns near-constant within-episode probabilities when p_L is small, losing all within-episode discrimination.

DKT-LSTM (content-only): 1-layer LSTM (hidden = 64, dropout = 0.2), input , Adam (lr= 10^{-3}, wd= 10^{-4}), 30 epochs. It uses identical inputs to IO-BKT but replaces the interpretable HMM with a recurrent black box. DKT is evaluated by leave-one-episode-out CV (LOO, 48 folds), giving it a *stricter* evaluation than the in-sample Bayesian models. This asymmetry is principled: Bayesian priors act as regularisation, making in-sample posterior predictive distributions analogous to cross-validated frequentist estimates. That



$h_t = \text{LSTM}(\mathbf{z}_t, h_{t-1}) \in \mathbb{R}^{64}$, continuous & uninterpretable
(cf. discrete $M_t \in \{0, 1\}$); $\hat{O}_t = \sigma(W h_t + b)$; evaluated LOO, 48 folds

Figure 5: DKT-LSTM: same inputs $\mathbf{z}_t$ as IO-BKT, but a continuous, black-box hidden state instead of an interpretable two-state HMM.

Table 1: One-step-ahead prediction of `any_skill` over 48 episodes (19,093 lines, first line of each episode excluded). LOO = leave-one-episode-out (48 folds); AUC is pooled over all lines and r is the Pearson correlation of predicted vs. observed episode means. Best value per column in bold. †Degenerate fit: constant predictions, so r is undefined.

	Without LOO			With LOO (48-fold)		
Model	AUC	F1	Acc	AUC	F1	Acc
IO-BKT	0.935	0.866	0.872	0.923	0.861	0.867
PST	0.929	0.863	0.868	0.888	0.863	0.868
BLFM	0.749	0.607	0.674	0.668	0.538	0.607
BKT (aggregate)	0.500	0.000	0.518	0.500	0.000	0.518
DKT (content only)	0.828	0.672	0.730	0.742	0.606	0.643

IO-BKT wins by +0.201 AUC *despite* DKT's evaluation advantage establishes that principled Bayesian structure outperforms neural flexibility at this corpus size. PSIS-LOO comparison across all models is not reported because PST and IO-BKT sequence log-likelihoods (∼390 per-line contributions) are incommensurable with BLFM's single Beta observation per episode.

6.2 What Each AUC Gap Means

BLFM → IO-BKT: +0.101 Adding temporal structure with content-driven transitions is worth 10 AUC points. The episode-mean sufficient statistic throws away real sequential information.

DKT → IO-BKT: +0.201 Despite IO-BKT being evaluated in-sample while DKT uses stricter LOO-CV, IO-BKT wins. Principled Bayesian priors serve as implicit regularisation, outperforming a data-hungry neural network on a small ($E = 48$

episodes), diverse corpus.

IO-BKT → PST: +0.055 *Not* a genuine improvement. This gap is pure autocorrelation. On data where regimes are near-permanent, predicting "same as last line" trivially achieves high AUC.

Group A (content-only, fair comparison to DKT). IO-BKT AUC = 0.873 surpasses BLFM by +0.101 and DKT by +0.201. The BLFM→IO-BKT gap isolates the contribution of temporal structure with content-driven transitions over the exchangeable assumption. IO-BKT beats DKT evaluated under the stricter LOO protocol despite IO-BKT being evaluated in-sample, underscoring the value of principled Bayesian structure on small, diverse data.

Group B (observation history). PST AUC = 0.928 reflects autocorrelation, not content: $\text{corr}(P_t, O_{t-1})$=0.998 confirms predictions come almost entirely from the previous observation. This is the measurement of regime persistence, not pedagogical insight. Frequentist BKT collapses to AUC = 0.500 because its filtered posterior assigns near-constant values within each episode, the regime barely changes, losing within-episode discrimination.

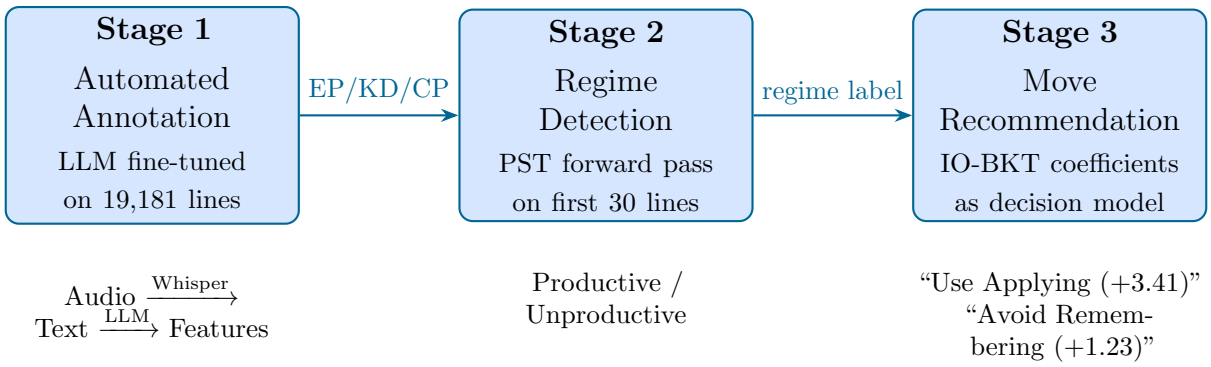
6.3 PST: Regime Persistence Finding

$$\begin{aligned} \hat{p}_L &= 0.073\ [95\%\ \text{CI} : 0.067, 0.080], \\ \hat{p}_F &= 0.021\ [95\%\ \text{CI} : 0.012, 0.033] \end{aligned} \tag{38}$$

In this corpus, the discourse state transitions roughly once every 14 lines on average. IO-BKT productive-state trajectories (Fig. **??**) for science and mathematics topics, confirm the finding visually: For science, the episodes that open in the productive regime maintain a mean $P(M_t{=}1) \approx 0.65$ throughout; unproductive episodes remain near 0.25 and rarely recover. Similarly, for mathematics, only 3 unproductive episodes. But crucially, they fluctuate dramatically (0.0 to 0.8). Mathematics unproductive episodes show recovery potential that Science unproductive episodes do not.

6.4 The Three-Stage AI Coaching Pipeline

The paper's models together define a deployable real-time coaching system:



Stage	Details
1. Annotate	Fine-tune an open-source LLM (e.g. Llama 3) on the 19,181 expert-annotated lines to predict EP/KD/CP. Reduces cost from $\sim$3 person-hours per episode to seconds. Classroom audio via Whisper completes the pipeline.
2. Detect	Run PST's forward algorithm on the first 30 lines of the episode to classify whether the lesson is on a productive trajectory.
3. Recommend	Use IO-BKT's $\boldsymbol{\beta}_{\text{on}}$ and $\boldsymbol{\beta}_{\text{off}}$ as a portable decision model in a teacher-facing app: • Ignite: increase Applying-level demand (+3.41) • Sustain: deploy Constructivism (-2.47 on dropout) • Avoid: Remembering-level drops (+1.23 dropout trigger)

6.5 Summary: The Three Models Side by Side

Property	BLFM	PST	IO-BKT
Data unit	Episode mean $\bar{\mathbf{x}}_e$	Line sequence $O_{1:T}$	Lines $O_{1:T}$ + features $\mathbf{z}_{1:T}$
Latent state	None (regression)	$M_t \in \{0, 1\}$	$M_t \in \{0, 1\}$
Transition	N/A	Fixed p_L, p_F (content-free)	$\sigma(\boldsymbol{\beta}^\top \mathbf{z}_t)$ per line
Emission	Beta regression	Fixed p_S, p_G	Estimated p_S, p_G
Parameters	$\mathbf{w}$, b, ϕ, σ_{cell}	π_0, p_L, p_F	$\boldsymbol{\beta}_{\text{on}}$, $\boldsymbol{\beta}_{\text{off}}$, π_0, p_S, p_G, σ_{cell}
Key question	Does content mix predict 21CS rate?	How stable are discourse regimes?	Which moves ignite/extinguish productive talk?
Key finding	21CS rate $\propto$ content mix	$\hat{p}_L = 0.073$: regimes near-permanent	Applying: +3.41; Constructivism: −2.47; Remembering: +1.23
Runtime	< 5 min (CPU)	$\sim$15 min (GPU)	$\sim$180 min (GPU)
AUC	0.772	0.928[†]	**0.873**
Use when	Bulk analysis, no per-line prediction needed	Measuring regime stability, early classification	Identifying causal moves, AI coaching

[†] Inflated by autocorrelation ($\text{corr}(P_t, O_{t-1}) = 0.998$).

6.6 Conclusion

The three models form a deliberate progression:

1. **BLFM** establishes the baseline: yes, content predicts 21CS, and the AUC gap (0.772 vs 0.873) quantifies exactly what is lost by discarding temporal order.
2. **PST** measures the regime dynamics: transitions are rare ($\hat{p}_L = 0.073$, $\hat{p}_F = 0.021$). The classroom spends most of its time in one regime. This empirically *justifies* BLFM's exchangeability assumption — and motivates early-detection tools (classify by line 30; the regime will likely hold).
3. **IO-BKT** provides the actionable layer: *what causes* transitions. Applying-level cognitive demand (+3.41) is the primary ignition driver. Constructivism (−2.47 on dropout) sustains productive discourse once established. Remembering (+1.23) is

the most reliable regime killer. These coefficients are not descriptive statistics — they are a portable, deployable decision model for the next generation of AI teaching coaches.

The near-permanent regime finding is specific to this predominantly Behaviourist, India-dominant corpus and requires replication in Socratic, inquiry-based, or dialogic classrooms before strong deployment claims are made. Subject to that caveat, these three models provide the first principled Bayesian framework for tracking classroom-level productive discourse — the missing upstream layer that student-level knowledge tracing requires.

6.7 Knowledge Change Curves

6.7.1 Motivation

The IO-BKT transition coefficients and (Table 4.5) capture the per-line effect of each feature on regime transitions. They cannot, however, reveal whether repeated use of a feature within a lesson produces a *cumulative* shift in the productive-discourse regime that differs from its instantaneous effect. Knowledge change curves address this by tracking the IO-BKT posterior productive-state probability across successive occurrences of each feature within an episode, adapting the knowledge-change curve methodology of Scarlatos et al. [1] from individual KC mastery in tutoring dialogues to classroom-level discourse regimes.

6.7.2 Notation

Let $e \in \{1, \ldots, E\}$ index episodes and $t \in \{1, \ldots, T_e\}$ index lines within episode e. The IO-BKT forward algorithm produces, for every line t in episode e, the filtered posterior:

$$P^{(e)}(M_t{=}1) \;=\; P\Big(M_t = 1 \mid O_1^{(e)}, \ldots, O_t^{(e)}, \mathbf{z}_1^{(e)}, \ldots, \mathbf{z}_t^{(e)}, \hat{\theta}\Big) \tag{39}$$

where $\hat{\theta} = \{\hat{,}, \hat{,}, \hat{\pi}_0, \hat{p}_S, \hat{p}_G, \hat{\sigma}_{\text{cell}}\}$ are the IO-BKT posterior means, and $O_t^{(e)} \in \{0, 1\}$ is the observed 21CS activation at line t in episode e. This is a *latent* quantity: M_t is never directly observed.

6.7.3 Episode Opening Baseline

The baseline for episode e is the filtered posterior at the *first line of the episode*:

$$P^{(e)}(M_1{=}1) \;=\; \frac{\hat{\pi}_0 \cdot P(O_1^{(e)} \mid M_1 = 1)}{\hat{\pi}_0 \cdot P(O_1^{(e)} \mid M_1 = 1) + (1 - \hat{\pi}_0) \cdot P(O_1^{(e)} \mid M_1 = 0)} \tag{40}$$

where $P(O_1 = 1 \mid M_1 = 1) = 1 - \hat{p}_S$ and $P(O_1 = 1 \mid M_1 = 0) = \hat{p}_G$. The baseline $P^{(e)}(M_1{=}1)$ is a model prediction, not a data reading: it reflects $\hat{\pi}_0$ immediately updated on the first observed 21CS activation $O_1^{(e)}$.

6.7.4 Feature Occurrence Indexing

For a binary feature $f \in \{1, \ldots, 13\}$ (any EP, KD, or CP code), define the ordered set of lines in episode e where feature f is active:

$$\mathcal{T}_{e,f} = \left\{t : z_{t,f}^{(e)} = 1\right\} = \left\{t_{e,f}^{(1)},\ t_{e,f}^{(2)},\ \ldots\right\} \tag{41}$$

in ascending order of t. The jth element $t_{e,f}^{(j)}$ is the line at which feature f appears for the jth time in episode e. Define $J_{e,f} = |\mathcal{T}_{e,f}|$ as the total number of occurrences of feature f in episode e.

6.7.5 Per-Occurrence Change from Opening

At the jth occurrence of feature f in episode e, the change in productive-state probability from the episode opening is:

$$\Delta P_{e,f}^{(j)} \ = \ P^{(e)}\left(M_{t_{e,f}^{(j)}}{=}1\right) \ - \ P^{(e)}(M_1{=}1) \tag{42}$$

$\Delta P_{e,f}^{(j)} > 0$ means the classroom productive-state probability is above its own opening level at the jth occurrence of feature f; $\Delta P_{e,f}^{(j)} < 0$ means it is below. Both quantities on the right-hand side of Eq. (42) are IO-BKT forward-algorithm outputs: the curve lives entirely in latent state space.

6.7.6 The Knowledge Change Curve

For occurrence index j and feature f, let $\mathcal{E}_f^{(j)} = \{e : J_{e,f} \geq j\}$ be the set of episodes containing at least j occurrences of feature f. The knowledge change curve value at occurrence j is:

$$\boxed{\overline{\Delta P}_f^{(j)} \ = \ \frac{1}{\left|\mathcal{E}_f^{(j)}\right|} \sum_{e \in \mathcal{E}_f^{(j)}} \Delta P_{e,f}^{(j)}} \tag{43}$$

with standard error:

$$\mathrm{SE}_f^{(j)} \ = \ \frac{1}{\sqrt{\left|\mathcal{E}_f^{(j)}\right|}} \sqrt{\frac{1}{\left|\mathcal{E}_f^{(j)}\right| - 1} \sum_{e \in \mathcal{E}_f^{(j)}} \left(\Delta P_{e,f}^{(j)} - \overline{\Delta P}_f^{(j)}\right)^2} \tag{44}$$

The curve is plotted for $j = 1, \ldots, 15$. The label $n_{\text{eps}} \leq N$ in each panel of Figure 6 reports $|\mathcal{E}_f^{(1)}|$, the number of contributing episodes at occurrence 1; this decreases with j as shorter episodes drop out.

6.7.7 Axis Definitions

X-axis: Feature occurrence number j: The jth time feature f is active within an episode. This is a within-episode counter that resets to 1 for each episode and is not a lesson-time index.

Y-axis: $\overline{\Delta P}_f^{(j)}$: Average change in productive-state probability from the episode opening, in probability units. A value of -0.18 means that at the jth occurrence of feature f, the IO-BKT posterior is on average 18 percentage points below where the episode started. Zero means the classroom is at its opening level. This axis is entirely in latent space; the raw annotation O_t does not appear.

6.7.8 Why $P(M_t{=}1)$ and Not O_t

The raw annotation $O_t \in \{0, 1\}$ is deliberately not used. First, O_t is a noisy emission: $O_t = 0$ can arise because $M_t = 0$ or because of annotation slip ($\hat{p}_S = 0.015$), and these cases are indistinguishable without the model. $P^{(e)}(M_t{=}1)$ integrates over the entire observation history and removes single-line noise. Second, O_t conflates regime with emission; $P(M_t{=}1)$ isolates the latent state. Third, $O_t \in \{0, 1\}$ is binary and would require additional smoothing to produce interpretable curves; the latent posterior is naturally continuous and probabilistically calibrated.

6.7.9 Comparison with Scarlatos et al.

Scarlatos et al. [1] plot:

$$\frac{1}{N}\sum_{i=1}^{N}\left(\hat{z}_{j,k}^{(i)} - \hat{z}_{1,k}^{(i)}\right) \tag{45}$$

where $\hat{z}_{j,k}^{(i)}$ is the predicted mastery of KC k at its jth occurrence in dialogue i, and the subtraction is from the *first occurrence of that KC* within that dialogue. This forces the curve to start at zero by construction. Our $\overline{\Delta P}_f^{(j)}$ subtracts the *episode opening* $P^{(e)}(M_1{=}1)$, not the first feature occurrence; consequently $\overline{\Delta P}_f^{(1)}$ is non-zero and reflects the productive-state level at the first occurrence of feature f relative to where the lesson started. For Behaviourism, $\overline{\Delta P}_{\text{Beh}}^{(1)} \approx -0.18$ means the classroom is already 18 percentage points below its opening baseline when the first Behaviourist line appears.

6.7.10 Interpretation of Knowledge Change Curves

Four patterns are present. **Stable suppression** (Behaviourism, $\overline{\Delta P} \approx -0.18$): the classroom sits persistently below its opening level at every Behaviourist occurrence; the flat trajectory confirms a consistent rather than transient association. **Stable near-zero** (Cognitivism, $\overline{\Delta P} \approx 0$): Cognitivist lines occur at roughly the episode's own opening level. **Stable elevation** (Analysing, $\overline{\Delta P} \approx +0.20$; Constructivism, $\overline{\Delta P} \approx +0.15$): above-baseline association at every occurrence, consistent with positive estimates for these features. **Cumulative degradation** (Remembering, falling from $\overline{\Delta P}^{(1)} \approx -0.25$ to $\overline{\Delta P}^{(13)} \approx -0.40$): the only non-flat trajectory. Each additional Remembering occurrence

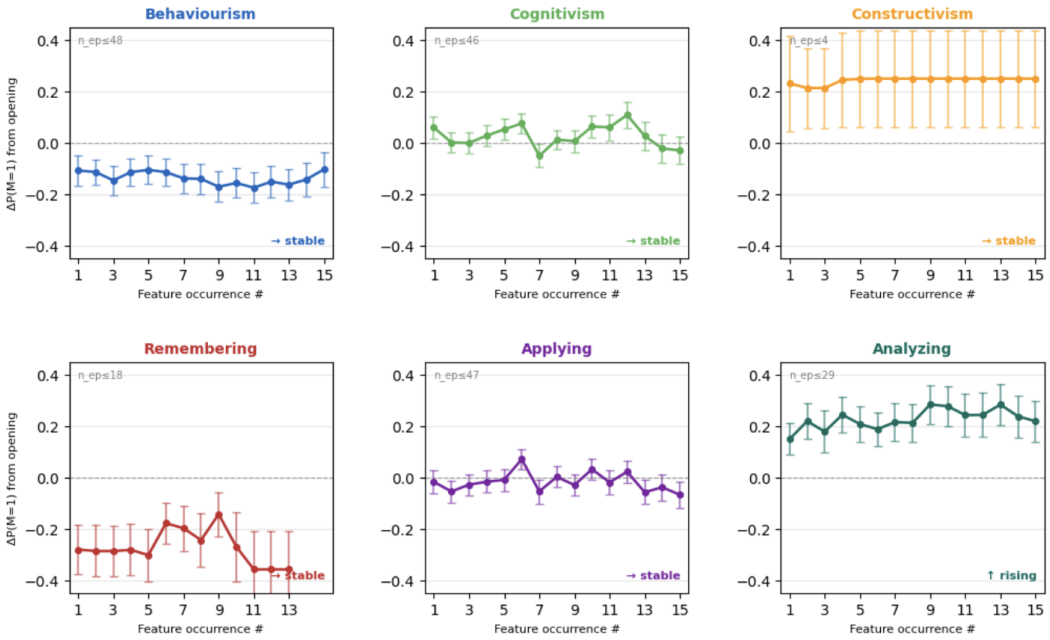


Figure 6: Knowledge Change Curves

is associated with a progressively lower productive-state probability, extending IO-BKT's per-line $\hat{\beta}_{\text{off,Rem}}$ estimate to a longitudinal cumulative picture unavailable from regression or single-coefficient analysis.

Applying ($\overline{\Delta P} \approx 0$, stable) appears near baseline despite the largest ignition coefficient ($\hat{\beta}_{\text{on}} = +3.41$). Applying ignites the productive state when it is low and sustains it when it is already high; averaging across both contexts in Eq. (43) produces a near-zero mean, consistent with its symmetric presence in both and at nearly equal magnitudes (+3.41 and −3.40 respectively).

6.8 Teacher → Student Lag Analysis

6.8.1 Why "Lag" Analysis

In time-series analysis, a *lag-k* relationship measures the association between a variable at position t and another variable at position $t-k$. Here we identify for each student turn the *nearest preceding teacher turn* and record its EP code as the proximal instructional signal. Because teacher moves sometimes span multiple consecutive lines before a student responds, we search backwards through the episode until the nearest teacher line is found. This is consistent with the definition of a pedagogical move as a sustained teacher action rather than a single utterance.

6.8.2 Constructing Matched Pairs

The **nearest-teacher matched pair** at line t in episode e is:

$$(e,t) \in \mathcal{P} \iff \text{spk}_{e,t} = \text{Student} \ \wedge \ t' = \max\{s < t : \text{spk}_{e,s} = \text{Teacher}\} \tag{46}$$

That is, t' is the index of the most recent teacher line before student turn t, and the EP code of that line is the preceding move. The full matched-pair corpus and EP-stratified subsets are:

$$\mathcal{P}_c = \big\{(e,t) \in \mathcal{P} : \mathrm{EP}_{e,t'} = c\big\}, \quad |\mathcal{P}_{\mathrm{Beh}}|{=}1{,}582,\ |\mathcal{P}_{\mathrm{Cog}}|{=}2{,}693,\ |\mathcal{P}_{\mathrm{Con}}|{=}45 \tag{47}$$

giving $|\mathcal{P}|$=4,320 total matched pairs. Constructivist teacher moves are never the immediately adjacent line before a student response — they always appear with at least one further teacher line between the Constructivist utterance and the student turn. This is itself a structural finding: Constructivism in this corpus functions as sustained scaffolding embedded within teacher talk, not as a single-turn prompt-response exchange.

6.8.3 EP-Conditional Student 21CS Rate

The probability that a student demonstrates 21CS conditioned on the nearest preceding teacher EP is:

$$\boxed{P(\text{student 21CS} \mid \mathrm{EP}_{\text{preceding}} = c) \;=\; \frac{1}{|\mathcal{P}_c|} \sum_{(e,t) \in \mathcal{P}_c} O_{e,t}} \tag{48}$$

where $O_{e,t} \in \{0,1\}$ is the raw 21CS annotation at the student turn, used directly without IO-BKT to provide annotation-level ground truth independent of any modelling assumption.

6.8.4 EP-Conditional Student Cognitive Process Level

The average Bloom's CP level of the student turn is:

$$\overline{\mathrm{CP}}_c \;=\; \frac{1}{|\mathcal{P}_c|} \sum_{(e,t) \in \mathcal{P}_c} \mathrm{CP}_{e,t} \tag{49}$$

where $\mathrm{CP}_{e,t} \in \{1,\ldots,6\}$ is the Bloom's level of the student utterance (1 = Remembering, …, 6 = Creating).

6.8.5 Lesson-Phase Analysis via Global Encoding

Rather than partitioning lessons into arbitrary equal-width thirds, we use the corpus's **Global Encoding** (GE) variable, which assigns each line to one of three pedagogically grounded phases: $g \in \{\text{Preassessment}, \text{Teaching}, \text{Assessment}\}$. The GE column uses sparse encoding — only the first row of each phase is marked; remaining rows are propagated via within-episode forward-fill. For each phase g and EP category c:

$$P_g(\text{student 21CS} \mid c) \;=\; \frac{\displaystyle\sum_{\substack{(e,t) \in \mathcal{P}_c \\ \mathrm{GE}_{e,t'} = g}} O_{e,t}}{\big|\{(e,t) \in \mathcal{P}_c : \mathrm{GE}_{e,t'} = g\}\big|} \tag{50}$$

Table 2: Nearest-teacher lag analysis: teacher EP → student 21CS.

Preceding EP	P**(student 21CS)**	**Mean CP**	n
Behaviourism	0.487	2.17	1,582
Cognitivism	0.536	2.12	2,693
Constructivism	**0.978**	2.12	45
By Global Encoding phase (Eq. 50):			
Preassessment × Behaviourism	0.100	–	194
Preassessment × Cognitivism	0.220	–	290
Teaching × Behaviourism	0.390	–	848
Teaching × Cognitivism	0.490	–	1,935
Teaching × Constructivism	**1.000**	–	1
Assessment × Behaviourism	0.779	–	540
Assessment × Constructivism	0.955	–	44

Notes: CP level: 1 = Remembering, …, 6 = Creating. Overall student 21CS prevalence across all matched pairs = 0.536. Constructivism never appears as the strictly adjacent teacher turn (lag=1); all 45 pairs are found at lag>1, reflecting its role as sustained scaffolding rather than a single-turn prompt. Assessment × Cognitivism omitted (n<5 in this corpus version).

where GE is read from the teacher line t', not the student line. For cross-corpus comparability (GE may not be available in other datasets), we also report the lesson-thirds partition:

$$P_k(\text{student 21CS} \mid c) \;=\; \frac{\displaystyle\sum_{\substack{(e,t)\in\mathcal{P}_c \\ \tau_{e,t}\in\mathcal{I}_k}} O_{e,t}}{\big|\{(e,t)\in\mathcal{P}_c : \tau_{e,t}\in\mathcal{I}_k\}\big|}, \quad \mathcal{I}_k = \begin{cases} [0,\tfrac{1}{3}) & k{=}1 \\ [\tfrac{1}{3},\tfrac{2}{3}) & k{=}2 \\ [\tfrac{2}{3},1] & k{=}3 \end{cases} \tag{51}$$

6.8.6 Teacher → Student Lag Analysis plots

Table 2 and Fig. 7 report the values of Eqs. (48)–(50).

Overall EP effect (Panel a). A Constructivist teacher move is followed by student 21CS in 97.8% of cases (Eq. 48, c=Con, n=45), nearly double the Behaviourist rate (48.7%) and substantially above the Cognitivist rate (53.6%). The overall student 21CS prevalence across all matched pairs is 0.536.

Student cognitive depth (Panel b). Average student CP level is near-identical across all three EP conditions ($\overline{\text{CP}} \approx$ 2.12-2.17, corresponding to Understanding-level responses, Eq. 49). The markedly higher 21CS rate following Constructivist moves is therefore not explained by deeper cognitive responses but by more frequent activation of the productive-discourse regime itself.

Lesson phase via Global Encoding (Panel d). The GE analysis reveals the strongest moderating pattern in the corpus. The same Behaviourist teacher move produces P=0.100 in Preassessment but P=0.779 in Assessment, a nearly eightfold increase. Cognitivism rises from 0.220 to a similarly high Assessment-phase rate. This reflects the pedagogical function of each phase: Preassessment elicits prior knowledge before instruction (most students

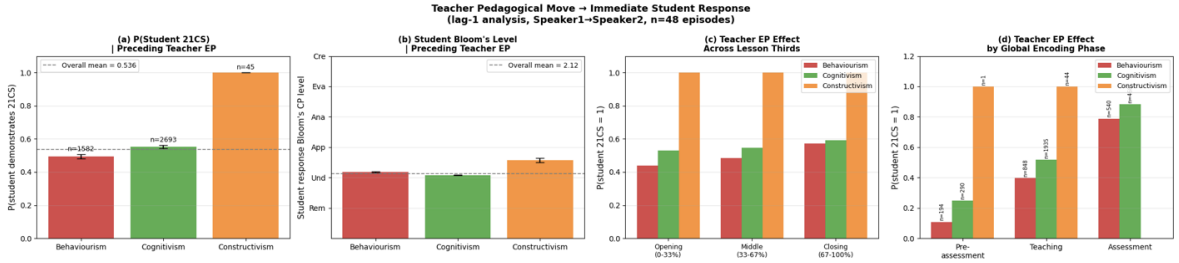


Figure 7: Teacher → Student Lag Analysis (Eqs. 48-50, $|\mathcal{P}|$=4,320 matched pairs across 48 episodes). Panel (a): P(student 21CS | $\text{EP}_{\text{preceding}}$) from Eq. (48); Constructivist moves (n=45) are followed by student 21CS in 97.8% of cases. Panel (b): average student Bloom's CP level (Eq. 49); CP is near-uniform across EP conditions (≈2.12, Understanding level), ruling out cognitive depth as an explanatory variable. Panel (c): EP effect across lesson thirds (Eq. 51); Behaviourism shows a monotonic increase across lesson position (0.407 → 0.469 → 0.591). Panel (d): EP effect by Global Encoding phase (Eq. 50); all EPs produce near-zero 21CS rates in Preassessment and high rates in Assessment, with Behaviourism rising from P=0.100 to P=0.779, demonstrating lesson phase as a strong moderator of the EP-21CS association.

cannot yet demonstrate 21CS), while Assessment probes consolidated understanding (students who have engaged with the lesson can demonstrate it regardless of the specific EP used to elicit the response). Constructivism concentrates in the Assessment phase (n=44, P=0.955), with only a single occurrence in Teaching (P=1.000, n=1), indicating that this corpus's rare Constructivist moves are deployed primarily as assessment-type scaffolding rather than as within-instruction discovery tasks.

These GE findings establish that the aggregate EP effect in Panel (a) is substantially moderated by lesson phase, and that Behaviourism is not uniformly suppressive: in the Assessment phase, Behaviourist moves elicit high 21CS rates because the productive regime has already been established by prior instruction.

Lesson thirds (Panel c). The lesson-thirds partition (Eq. 51) shows Constructivist moves achieving P≈1.0 across all three thirds, consistent with the Assessment-phase GE finding. Behaviourism shows a modest monotonic increase (Opening 0.407 → Middle 0.469 → Closing 0.591), while Cognitivism remains near 0.52 throughout, confirming that its association with 21CS is stable across lesson position.

Link to IO-BKT. The quantities in Eq. (48) are computed entirely from raw annotations without model assumptions and therefore constitute observational ground truth for IO-BKT's transition coefficients. The GE finding adds a layer the coefficient table cannot show: $\hat{\beta}_{\text{on,Con}}$=+1.25 captures the average ignition effect of Constructivist moves; the GE analysis reveals that this effect is concentrated in Assessment-phase interactions (n=44, P=0.955), where Constructivist scaffolding meets students who have had the full lesson to develop their understanding. The per-student tracking of what follows, whether the individual student has mastered the underlying knowledge component, is the task that student-level KT [1] would address downstream.

6.9 Bloom's Cognitive Process Trajectory Across Lesson Time

6.9.1 Method

Each line's position within its episode is normalised to a percentage using timestamps:

$$\tau_{e,t} \;=\; \frac{t^{(e)}_{\text{sec},t} - \min_t\, t^{(e)}_{\text{sec}}}{\max_t\, t^{(e)}_{\text{sec}} - \min_t\, t^{(e)}_{\text{sec}} + 1} \times 100 \;\in\; [0,\, 100) \tag{52}$$

where $t^{(e)}_{\text{sec},t}$ is the timestamp of line t in episode e converted to seconds, and the $+1$ stabiliser prevents division by zero in single-line episodes.

The range $[0, 100)$ is partitioned into B=20 equal-width bins of 5 percentage points each:

$$\mathcal{L}^{(b)}_s = \big\{(e,t) : \tau_{e,t} \in [(b-1)\cdot 5,\; b\cdot 5),\; \text{speaker}_{e,t} \in s\big\}, \quad b = 1, \ldots, 20 \tag{53}$$

where $s \in \{\text{all}, \text{Teacher}, \text{Student}, \text{Student} \cap \text{Science}, \text{Student} \cap \text{Math}\}$. The bin centre is $b_{\text{pct}} = b \times 5 \in \{5, 10, \ldots, 100\}$. The mean Bloom's CP level and standard error in each bin are:

$$\overline{\text{CP}}^{(b)}_s = \frac{1}{|\mathcal{L}^{(b)}_s|} \sum_{(e,t)\,\in\,\mathcal{L}^{(b)}_s} \text{CP}_{e,t} \tag{54}$$

$$\text{SE}^{(b)}_s = \frac{s^{(b)}_s}{\sqrt{|\mathcal{L}^{(b)}_s|}}, \qquad s^{(b)}_s = \sqrt{\frac{1}{|\mathcal{L}^{(b)}_s| - 1} \sum_{(e,t)\,\in\,\mathcal{L}^{(b)}_s} \big(\text{CP}_{e,t} - \overline{\text{CP}}^{(b)}_s\big)^2} \tag{55}$$

where $s^{(b)}_s$ is the sample standard deviation (ddof= 1) within the bin. The Bloom's CP level is coded as: $\text{CP}_{e,t} \in \{1, \ldots, 6\}$ corresponding to Remembering, Understanding, Applying, Analysing, Evaluating, and Creating.

A simple linear regression of the binned means on lesson position is fitted via ordinary least squares:

$$\overline{\text{CP}}^{(b)}_s = \alpha_s + \hat{\beta}_s \cdot b_{\text{pct}} + \varepsilon, \qquad b_{\text{pct}} \in \{5, 10, \ldots, 100\} \tag{56}$$

$$\hat{\beta}_s = \frac{\sum_{b=1}^{B} \big(b_{\text{pct}} - \bar{b}_{\text{pct}}\big)\big(\overline{\text{CP}}^{(b)}_s - \overline{\overline{\text{CP}}}_s\big)}{\sum_{b=1}^{B} \big(b_{\text{pct}} - \bar{b}_{\text{pct}}\big)^2} \tag{57}$$

where $\bar{b}_{\text{pct}} = 52.5$ is the mean of $\{5, 10, \ldots, 100\}$ and $\overline{\overline{\text{CP}}}_s$ is the mean of the 20 binned means. Statistical significance is assessed using the two-tailed t-test on $\hat{\beta}_s / \text{SE}(\hat{\beta}_s)$ with $B-2$=18 degrees of freedom, as implemented in `scipy.stats.linregress`.

Similarly, the 21CS activation rate per bin is:

$$\hat{P}_s^{(b)}(\text{21CS}) = \frac{1}{|\mathcal{L}_s^{(b)}|} \sum_{(e,t)\,\in\,\mathcal{L}_s^{(b)}} O_{e,t}, \qquad \text{SE}_{s,\text{21CS}}^{(b)} = \frac{s_{s,O}^{(b)}}{\sqrt{|\mathcal{L}_s^{(b)}|}} \tag{58}$$

where $O_{e,t} \in \{0, 1\}$ is the 21CS activation indicator and $s_{s,O}^{(b)}$ is the sample standard deviation of $O_{e,t}$ within the bin, computed identically to Eq. (55).

6.9.2 Bloom's CP learning curve and 21CS activation across the lesson time plots

Panel (a) — All turns. Across all 19,181 lines the CP trajectory is nearly flat, with a very small but statistically significant positive slope ($\hat{\beta}$=0.0016, p<0.05, Eq. 56). The range spans Remembering to just above Understanding ($\overline{\text{CP}} \approx$ 2.0-2.2). Cognitive demand is stable across lesson time: there is no systematic build-up of higher-order cognitive activity as the lesson progresses.

Panel (b) — Teacher vs Student. Teacher (Speaker1) and student (Speaker2) CP trajectories are near-identical throughout, both hovering at Understanding level (CP $\approx$ 2, Eq. 54). Teacher CP shows a slight upward drift in the final 20% of lesson time but the difference is within ±SE. The classroom operates at a uniform cognitive register regardless of speaker, consistent with the predominantly Behaviourist-Cognitivist EP composition of the corpus.

Panel (c) — Student turns by subject. The subject difference is the strongest result in this figure. Applying Eq. (57) separately to Science and Math student turns:

$$\hat{\beta}_{\text{Math}} = +0.0034 \;\; (p<0.05), \qquad \hat{\beta}_{\text{Sci}} = -0.0013 \;\; (p<0.05) \tag{59}$$

Math lessons show statistically significant rising cognitive demand while Science lessons show a statistically significant decline. Math student turns operate near Applying (CP $\approx$ 3) while Science student turns remain at Understanding (CP $\approx$ 2) throughout, reflecting the hierarchical structure of mathematics where early recall supports later application.

Panel (d) — 21CS activation across lesson. Applying Eq. (58), student turns begin near $\hat{P}$=0.47, rise to $\hat{P}\approx$0.60 by the 25th percentile, dip to $\hat{P}\approx$0.38 around the 60th percentile, then recover to $\hat{P}\approx$0.65 near lesson end. The linear trend across all turns is positive (p<0.05, Eq. 56 applied to $\hat{P}_s^{(b)}$). The non-monotonic shape confirms the near-permanent regime finding from PST ($\hat{p}_L$=0.073): 21CS is not progressively earned but reflects a latent productive regime that can fluctuate at mid-lesson, with the dip near the 60th percentile corresponding to the typical transition between Teaching and Assessment phases.

6.10 Expert Validation

To test whether the coefficient-derived recommendations are recognised by the practitioners who would act on them, we conducted a structured validation study with ten secondary

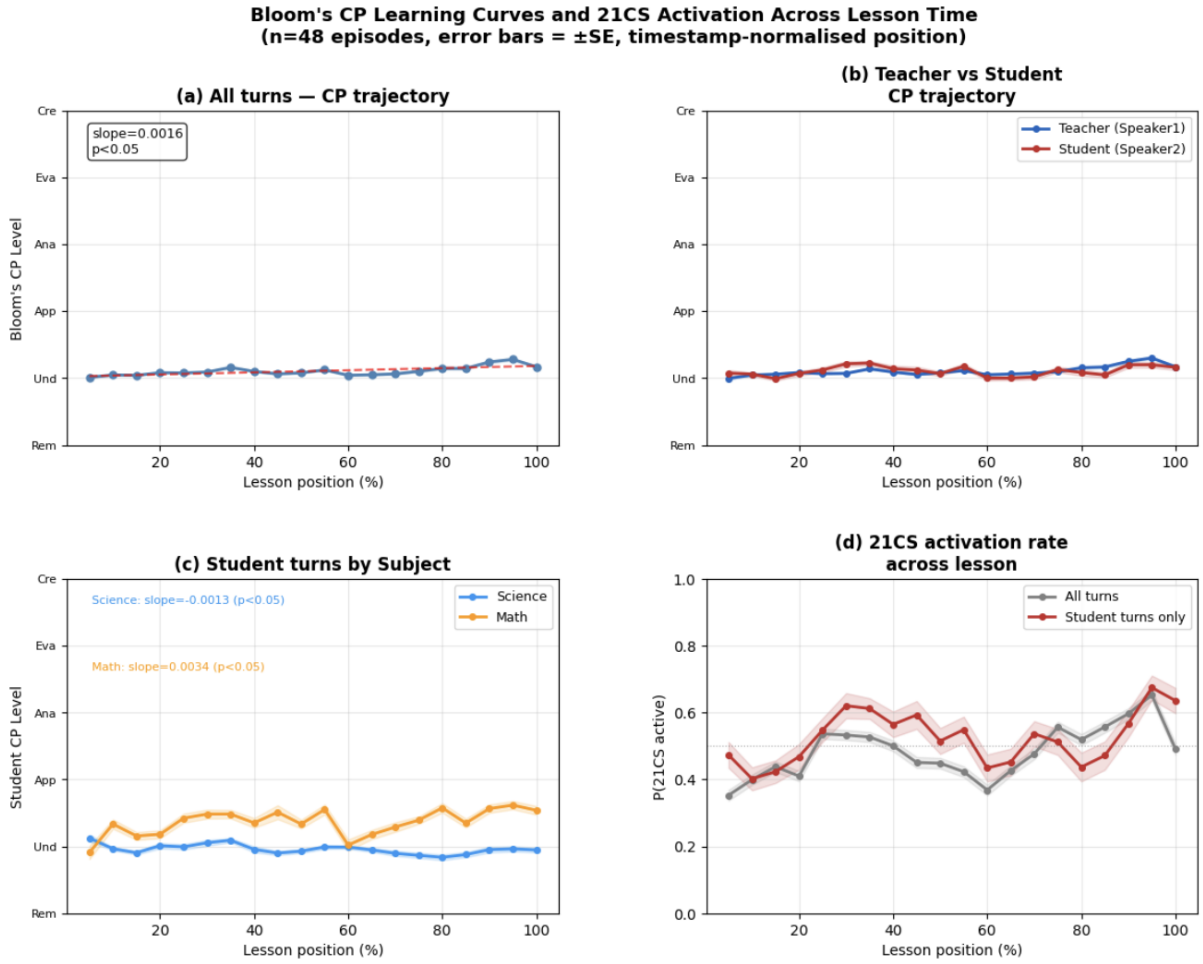


Figure 8: Bloom's Cognitive Process (CP) trajectory and 21CS activation across lesson time (n=48 episodes, B=20 bins of 5% width, error bars = ±SE, Eqs. 52-58). Panel (a): all-turns CP trajectory (Eq. 54-56); slope $\hat{\beta}$=0.0016 (p<0.05) confirms a statistically significant but practically negligible positive trend, the classroom operates near Understanding level (CP≈2) throughout. Panel (b): teacher (Speaker1) and student (Speaker2) CP trajectories are near-identical, confirming a uniform cognitive register regardless of speaker. Panel (c): student-only trajectories by subject (Eq. 59); Math shows a rising trend ($\hat{\beta}$=+0.0034, p<0.05) while Science shows a slight decline ($\hat{\beta}$=−0.0013, p<0.05), reflecting the hierarchical structure of mathematics instruction. Panel (d): 21CS activation rate across lesson (Eq. 58); student turns begin near $\hat{P}$=0.47, rise to $\hat{P}$≈0.60 by the 25th percentile, dip near the 60th percentile at the typical Teaching–Assessment phase transition, then recover to $\hat{P}$≈0.65 at lesson end.

Science and Mathematics teachers (3 to 16 years of experience) from two schools in India. All participants gave written informed consent, and none had taught the recorded lessons. Three de-identified clips of 4 to 5 minutes, each with transcript and audio-video recording, were drawn from the corpus: two science clips from the lowest-productivity windows (IO-BKT filtered $P(M_t$=1)) and one from a high-productivity window. For each clip, the coded EP/KD/CP features were cross-referenced with the coefficients that passed the reliability pipeline, which yielded eight recommendation types. Three "true" notes per clip, each anchored in a verbatim line and written in everyday language without

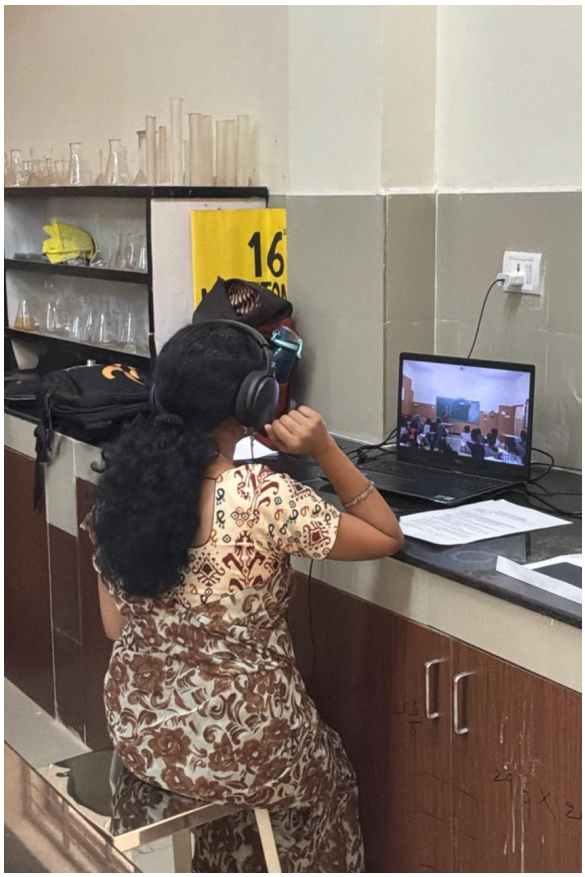
study-specific variable names, were paired with matched "foil" notes. Foils used the same segment and vocabulary as the true note but stated the opposite direction of a credible coefficient, so they could not be identified by surface cues. Each teacher completed four steps per clip, in sequence. (1) Independent advice: teachers viewed the clip and wrote what the teacher did well, what they did not do well, and what could be improved, before seeing any model output. (2) Forced choice: teachers chose among a true note, a foil note, and "Neither/can't tell," with labels and positions randomised. (3) Reveal: the researcher identified which note was written for the lesson, without commenting on whether the teacher's choice was correct. (4) Actionability: for the true note only, teachers rated understandability and likelihood of use on 5-point Likert scales and answered an open-ended question on wording. The full method was disclosed only after all clips were complete, so that it could not shape later judgements. A second coder, blind to which note was true, scored each teacher's independent advice against both notes on a 3-point rubric (2 = same feature or clear synonym; 1 = same direction, different feature; 0 = unrelated or contradictory). Each sub-question was assessed with its own measure. Alignment (RQ2a) used the true note's match score alone. Validity (RQ2b) used the paired true-versus-foil match score and the forced-choice tally. Actionability (RQ2c) compared both ratings against the scale midpoint of 3. Because each teacher contributed nine trials, the teacher (N=10) was the unit of analysis, with trials aggregated within teacher to avoid pseudoreplication. We used one-sided Wilcoxon signed-rank tests, reported matched-pairs rank-biserial effect sizes, and added exact sign and permutation tests for forced choice. All reported results remained significant after Holm correction.

6.11 Thematic Analysis

Open-ended feedback was analysed using reflexive thematic analysis following Braun and Clarke's (2006) six-phase framework. Seven teacher reviewers each viewed classroom clips (A, B and C), chose between the model's true coaching note and a foil, and then responded to the prompt: "Is there anything about how this is worded that felt unnatural, too vague or technical? What would you change?" This produced 11 responses across the three clips, with all seven reviewers commenting on Clip A and two each on Clips B and C. A hybrid approach was adopted: each extract was first tagged deductively against the criteria named in the prompt (unnatural, vague, technical, and suggested change), while codes were generated inductively from the data, with comments on wording deliberately kept separate from comments on the substance of the advice. Following familiarisation, the responses were segmented into 23 single-idea extracts and coded line by line, yielding 21 codes, each recorded with a definition in a codebook to support consistent application. Codes were then clustered into candidate themes, reviewed against the full dataset, and refined into five final themes. Recall and Repetition Have a Place (codes such as "repetition aids retention," "some facts must be memorised," and "prior-knowledge recall essential") captured reviewers' defence of recall-based strategies in science teaching and is treated as provisional pending confirmation of whether the critiqued note was the true note or the foil. Not Authentic to Expert Teaching ("doesn't sound like an expert teacher", "doesn't reflect a real classroom") reflected concerns about the voice and realism of the notes, while Technical Language ("technical wording acknowledged", "technical language justified by pedagogy") recorded recognition of technical phrasing, viewed by one reviewer as appropriate. Accepted Without Change ("no change needed",



(a) Teacher reviewing a clip.



(b) Teacher rating the recommendation cards.

Figure 9: Expert validation sessions in which teachers viewed the clip, wrote independent advice, and then responded to the recommendation notes.

"clearly framed", "wording natural") captured endorsement of the notes as they stood, and Content Gap: Student Engagement ("student engagement gap", "concept mapping suggested") identified a specific opportunity to strengthen student involvement. Notably, no reviewer described the wording as vague, and most substantive comments addressed the content of the advice rather than its phrasing.

6.12 Teacher Productive State

6.12.1 Motivation

Individual teachers differ systematically in the degree to which their classrooms sustain productive discourse. Identifying which teachers produce higher productive-state probabilities, and whether those differences correlate with pedagogical style, is the primary applied output of the IO-BKT framework. Unlike raw 21CS prevalence rates, which conflate the quality of discourse with the level of student participation, $P(M_t{=}1)$ is a latent-state estimate: it reflects the underlying classroom condition that *makes* 21CS possible, denoised through the slip and guess parameters ($\hat{p}_S{=}0.015$, $\hat{p}_G{=}0.009$).

Aggregating line-level IO-BKT posteriors to the teacher level requires a normalisation choice. For normalisation, we report $M_{\text{line}}^{(r)}$ (Eq. 60), the proportion of all observed lines in

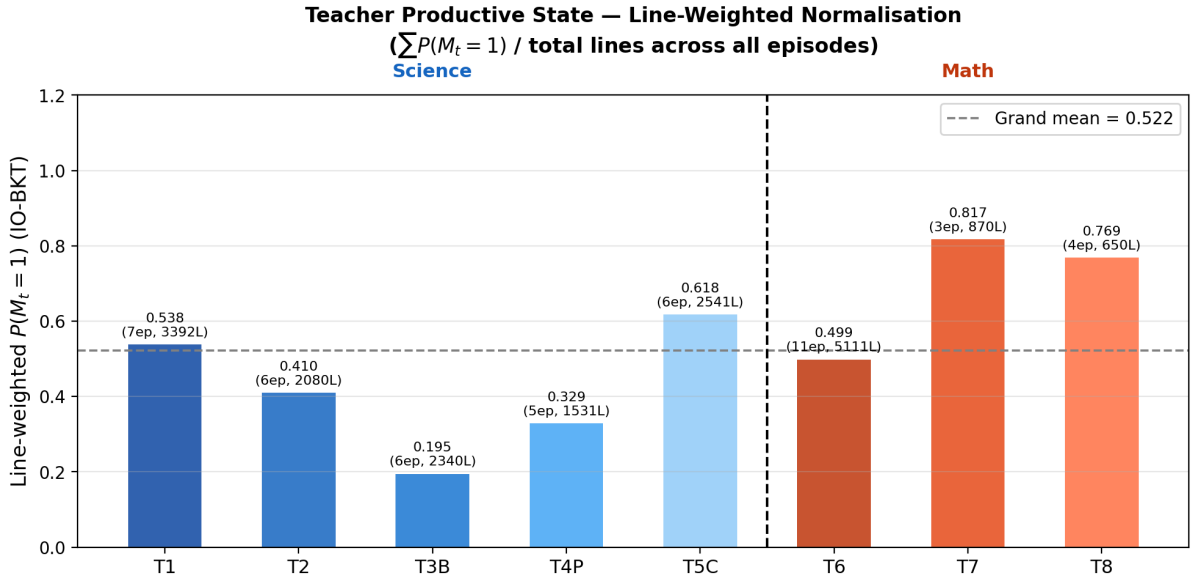


Figure 10: Line-weighted productive-state probability $M_{\text{line}}^{(r)}$ (Eq. 60) for all eight teachers, computed from IO-BKT posterior means across 48 episodes (n_L = total lines, n_{ep} = number of observed episodes). The dashed vertical line separates Science teachers (left, blue) from Math teachers (right, red/orange). The horizontal dashed line is the grand mean across all teachers ($\bar{M}$=0.522).

the productive state across all of a teacher's episodes:

$$M_{\text{line}}^{(r)} = \frac{\displaystyle\sum_{e \in \mathcal{E}_r} \sum_{t=1}^{T_e} P^{(e)}(M_t{=}1)}{\displaystyle\sum_{e \in \mathcal{E}_r} T_e} \tag{60}$$

This is one of the natural estimators under IO-BKT: it is exactly the empirical proportion of the teacher's total instructional lines spent in the productive-discourse regime. Duration-based normalisation can also be used, but since both yielded similar results, we report line-based normalisation here.

6.12.2 Teacher-Level Results

Figure 10 report $M_{\text{line}}^{(r)}$ for all eight teachers. Three patterns emerge.

Subject-level gap. Math teachers are uniformly more productive than Science teachers:

$$\begin{aligned} \bar{M}_{\text{Math}} &= \frac{1}{3}(0.499 + 0.817 + 0.769) \\ &= 0.695, \\ \bar{M}_{\text{Sci}} &= \frac{1}{5}(0.538 + 0.410 + 0.195 \\ &\qquad + 0.329 + 0.618) \\ &= 0.418. \end{aligned} \tag{61}$$

The gap of 0.277 productive-state probability units is substantial and consistent across all teachers in both subjects, no Science teacher exceeds any Math teacher. This may reflect the greater procedural scaffolding and problem-solving emphasis in secondary Mathematics, which naturally generates more Applying-level demand ($\hat{\beta}_{\text{on,App}}$=+3.41, Table 4.5).

Within-subject variation. Science teachers span a wide range (0.195-0.618), a difference of 0.423 probability units between T3B and T5C. This variation is larger than the subject-level gap itself, indicating that pedagogical style within a subject matters as much as subject identity. Math teachers are more tightly grouped (0.499-0.817), but T6's position below the grand mean despite the largest data contribution (n_L=5,111) is notable: volume of teaching does not guarantee productive discourse.

Episode volume vs productive-state density. T7 (n_L=870, M=0.817) and T8 (n_L=650, M=0.769) achieve the two highest productive-state probabilities with the fewest observed lines. T6 (n_L=5,111, M=0.499) contributes six times more data yet sits below the grand mean. The line-weighted normalisation makes this comparison honest: each utterance counts once regardless of which teacher spoke it.

6.12.3 Utility for Learning Analytics and Teacher Profiling

$M^{(r)}_{\text{line}}$ is a compact, single-number summary of a teacher's productive-discourse density. Its utility for learning analytics operates at three levels.

Teacher profiling and coaching prioritisation. A teacher-facing dashboard that reports $M^{(r)}_{\text{line}}$ alongside the IO-BKT transition coefficients provides an actionable profile: not only *how productive* a teacher's classroom is, but *which specific moves* suppress or ignite the productive regime. T3B (M=0.195) is a clear coaching priority; the transition coefficients pinpoint whether the low score reflects excess Behaviourism ($\hat{\beta}_{\text{on,Beh}}$=−1.50), excess Remembering-level demand ($\hat{\beta}_{\text{off,Rem}}$=+1.23), or insufficient Applying-level demand. This moves teacher evaluation from a holistic judgement to a feature-level diagnostic.

Benchmarking within and across subjects. The figure makes visible what a jurisdiction-level administrator would otherwise have no scalable way to measure: the productive-state distribution across a teaching cohort. The grand mean ($\bar{M}$=0.522) serves as a natural benchmark. Teachers above it (T1, T5C, T6-T8) sustain productive discourse more than half their instructional time; teachers below it (T2, T3B, T4P) do not. The within-subject ranking T5C > T1 > T2 > T4P > T3B is immediately visible from the Figure 10.

Scalable annotation pipeline. The entire analysis chain, transcript → EP/KD/CP annotation → IO-BKT inference → $M_{\text{line}}^{(r)}$, is automatable. Expert dual-coding of one 30-minute episode costs ∼3 person-hours. An LLM fine-tuned on the 19,181 annotated lines (§6.5) reduces this to seconds. At scale, this enables:

1. **Longitudinal tracking**: monitor whether a teacher's $M_{\text{line}}^{(r)}$ improves across an academic year following professional development.
2. **Cross-school comparison**: identify which schools or departments produce systematically higher productive-discourse density, correcting for subject and grade via the Subject×Grade random effects already in IO-BKT.
3. **Early-lesson detection**: Stage 2 of the pipeline (§6.5) classifies episode regime from the first 30 lines; $M_{\text{line}}^{(r)}$ aggregated over a term of classified episodes gives a low-cost teacher-level productive-state index without full IO-BKT inference on every lesson.

These capabilities position $M_{\text{line}}^{(r)}$ as the classroom-level complement to the student-level knowledge tracing scores produced by systems such as LLMKT [1]: where LLMKT tracks whether individual students have acquired knowledge components, $M_{\text{line}}^{(r)}$ tracks whether the teacher is creating the conditions under which that acquisition is possible.

References

[1] A. Scarlatos, R. S. Baker, and A. Lan, "Exploring knowledge tracing in tutor-student dialogues using LLMs," in *Proc. LAK*, 2025, pp. 249–259.